XPP Studio

An introduction and tutorial

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Abstract

XPP Studio is a browser-based environment for exploring dynamical systems, modelled on Bard Ermentrout's XPPAUT. It reads XPPAUT's `.ode` files and provides phase planes, three-dimensional views, time courses, numerical continuation of equilibria and periodic orbits, one- and two-parameter bifurcation diagrams, maps, stochastic and Markov models, stiff solvers, delay equations, events (global flags), boundary value problems, arrays and Volterra integral equations. Everything runs locally in a web page: there is nothing to install and no data leaves the browser. This document introduces the interface, describes the model language, and works through a sequence of short tutorials using the examples that come with the program.

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1 Introduction

For more than thirty years XPPAUT [1] has been one of the standard tools for simulating and analysing differential equations in mathematical biology, neuroscience and applied dynamical systems. Its strengths are a compact, readable model language, fast interactive phase-plane analysis, and a tight connection to the continuation package AUTO [2]. XPP Studio keeps that model language and that way of working, and puts them in a modern interface:

- **Same files.** Most XPPAUT `.ode` files load unchanged. Supported features include parameters, functions, fixed quantities, auxiliary variables, maps, Wiener noise, Markov variables, tables, periodic (folded) variables, arrays, Volterra integrals, delays, global flags and boundary conditions (Section 3).
- **Live analysis.** Dragging a parameter slider immediately recomputes trajectories, nullclines and equilibria.
- **Built-in continuation.** Equilibria, periodic orbits and their bifurcations can be continued in one or two parameters without leaving the program. Detected points include folds, Hopf, period-doubling, torus, fold-of-cycles and codimension-two points.
- **Publication output.** Every view can be exported as a vector SVG figure. Continuation data can be exported as CSV, and trajectories copied as CSV.

The program is a single web page. To use it, open

`https://xpp-studio.github.io/xpp-studio/`

in any recent browser. The source code, the example models and this tutorial are in the repository `https://github.com/xpp-studio/xpp-studio`. Its file `index.html` is the whole program, and it also runs when opened directly from disk.

How this document is organised. Section 2 is a tour of the interface and Section 3 summarises the model language. Sections 4–11 are hands-on tutorials, each built on an example from the **Model** menu. Section 15 shows how to record and replay the steps behind a set of figures with a script. Sections 16 and 17 are two further tutorials, on a forced oscillator and on homoclinic and heteroclinic orbits. Section 18 covers numerical settings and Section 19 exporting results. The document ends with the differences from XPPAUT and with credits.

2 A tour of the interface

The window (Figure 1) has three columns.

Left: the model. Five tabs:

Equations The `.ode` text, which you can edit freely. **Apply** (or `Ctrl`+`Enter`) reloads the model, and messages underneath report what was loaded and any errors, with line numbers. **Open .ode** loads a file from disk, **Copy text** copies the current text, **Write state** writes the current parameters, initial conditions and settings into it (Section 19), and **Table files** supplies data files used by `table` declarations.

Parameters One row per parameter, with a value box and a slider whose limits you can edit. Drag a slider to see everything update live, and use **Reset** to restore the values in the file. The up and down arrows on a value box, and the arrow keys on a slider, change the value by 1% of the slider's range (10% with `Shift`).

Initial Initial values of the state variables (and Markov states). **Integrate** runs from here, and **Last point** → **IC** copies the end of the current trajectory into the initial conditions.

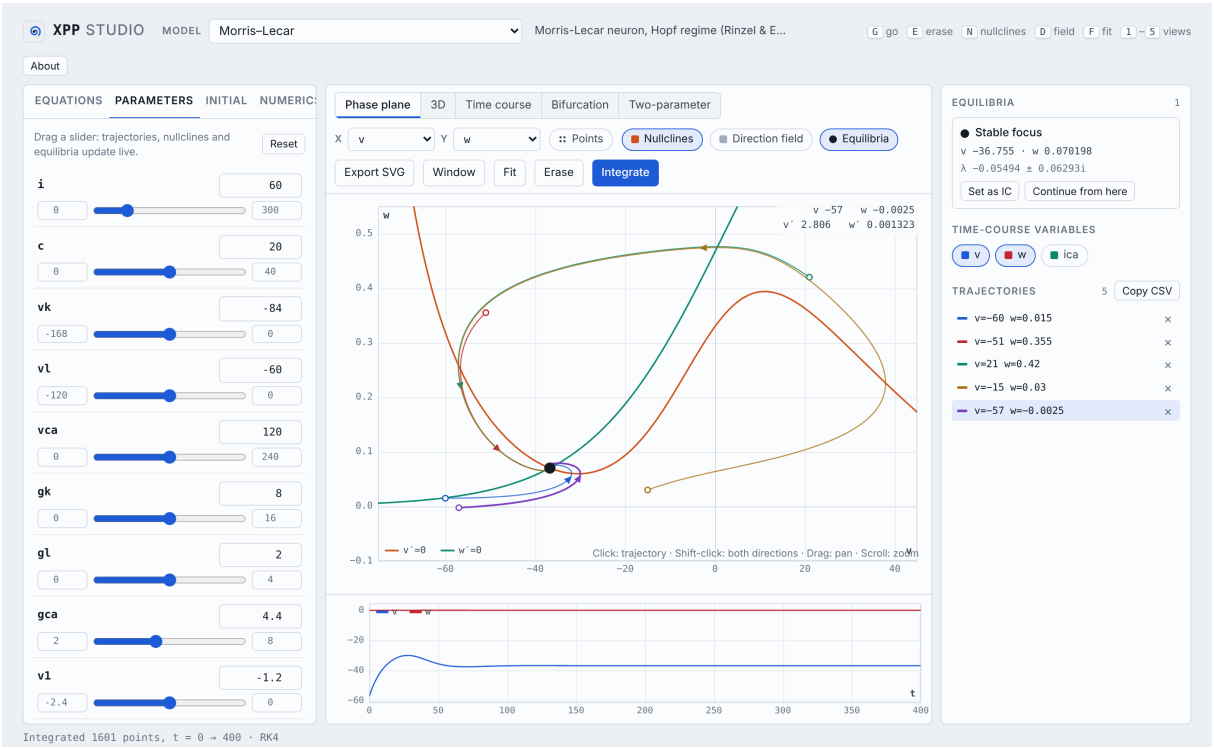


Figure 1: The XPP Studio window with the Morris–Lecar example loaded. Left: tabs for equations, parameters, initial conditions and numerics. Centre: the plot, here the (v, w) phase plane with nullclines, five trajectories and a stable focus, plus a time-course strip underneath. Right: the inspector, listing equilibria, the variables shown in the time course, and the trajectories.

Numerics Integration method, total time, transient, step size, tolerance and bound, plus the continuation settings (Section 18).

Script A list of commands that drives the interface, to make a set of figures again exactly, with one click (Section 15).

Centre: the plot. Five views, chosen with the buttons above the plot or with the keys 1–5:

Phase plane Any two variables (state, Markov, Volterra or auxiliary) against each other, with optional nullclines, direction field and equilibria. For a one-dimensional map this view shows a cobweb diagram. **Equal axes** gives both axes the same scale. **Run grid** in the Initial tab starts trajectories from a grid of points, by default covering the window (Section 4). Below the plane a strip shows the time series of the current trajectory.

3D Three variables (or time) in a rotatable box: drag to rotate, scroll to zoom, **Spin** to animate.

Time course Chosen variables against t (or the iteration number n for maps). **Spectrum** shows their power spectra instead, and **Space-time** shows a row of indexed variables as an image (Section 11).

Bifurcation Branches of equilibria and periodic orbits, parameter sweeps, and homoclinic and heteroclinic orbits, against one parameter. The vertical axis is a variable, or the period or frequency of the periodic orbits.

Two-parameter Curves of folds, Hopf points, period doublings and bifurcations of cycles in a parameter plane.

The status line at the bottom of the window reports what the last action did: how many points a trajectory has, why a branch stopped, where a special point was found. **History**, at

its right-hand end, lists every message since the page was opened, each with its time, so that a report that has scrolled away can be read again. Progress reports during a long computation are kept as one line, the last one. **Download** saves the list as a text file, **Copy** copies it and **Clear** empties it; in a script, `history save` does the same (Section 15).

In every view you drag to pan and scroll to zoom. **Fit** rescales to the data, and **Window** sets the axis limits by hand (they then stay fixed until you press **Fit**). **Points** draws trajectories as dots instead of lines, which is the default for maps. **Export SVG** saves the current view as a vector figure. **Export CSV** saves the data behind the time course, bifurcation diagram or two-parameter diagram, and **Export .dat** saves continuation data as plain columns of numbers (Section 19).

Labelling a plot. The **Note** chip above the plot puts a text label on the figure. It appears in the phase plane, the time course (including the spectrum and the space-time plot), the Bifurcation view and the two-parameter plane, but not in the 3D views. Press it, click the plot where the label should go, and type the text in the small box that opens; a new line in the box starts a new line of text. The box also sets the size (small, medium or large) and the colour: the ordinary text colour, the accent, one of the colours used for curves and trajectories, or any colour from the colour picker. Close the box with **Done**, Ctrl+Enter, Esc or a click elsewhere; a note with no text is discarded.

- **Move, edit, delete.** Drag a note to move it, double-click it to edit it, or click it to select it and press Delete. **Delete** in the editing box removes it too.
- **Notes belong to a plot.** A note is placed in the data coordinates of the plot it was made on, so it stays on its feature when you pan and zoom. It is shown only on that plot: the phase plane of that pair of variables, the diagram for that continuation parameter and vertical axis, or that parameter plane. Switch to other axes and the note is hidden, and it comes back when you return. Loading another model removes all notes.
- **In figures and scripts.** Notes are drawn in **Export SVG**, in the colours of the light theme, along with the legend. A note you make while recording is written into the script as a `note` line (Section 15). Notes are not saved in the `.ode` file.

Right: the inspector. The inspector's contents depend on the view. In the phase plane it lists the equilibria (type, coordinates and eigenvalues, with buttons to make one the initial condition, start continuation from it, or draw its separatrices), the variables in the time course, and the trajectories. For large systems, and for boundary value problems, equilibria are not searched for automatically; **Find** next to the list searches the visible window when you want them. In the Bifurcation view it holds the continuation controls, the search for homoclinic and heteroclinic orbits, and the table of special points. In the Two-parameter view it holds the two-parameter controls.

Key	Action
G	integrate from the current initial condition ("go")
E	erase all trajectories
N / D	toggle nullclines / direction field
F	fit the axes to the data
1–5	phase plane, 3D, time course, bifurcation, two-parameter
Esc	during continuation: skip to the other direction ("soft interrupt")
Shift+Esc	stop the continuation altogether
click / shift-click	start a trajectory (shift: in both directions of time)

Table 1: Keyboard shortcuts and mouse actions.

3 The model language

A model is a plain-text `.ode` file in XPPAUT's format. Lines starting with `#` are comments, and the first comment line becomes the model's title. Names are case-insensitive. The Morris–Lecar example [10, 11] illustrates most of the basic syntax:

```
# Morris-Lecar neuron, Hopf regime (Rinzel & Ermentrout)
dv/dt=(i+gl*(vl-v)+gk*w*(vk-v)+gca*minf(v)*(vca-v))/c
dw/dt=phi*(winf(v)-w)/tauw(v)
minf(v)=.5*(1+tanh((v-v1)/v2))
winf(v)=.5*(1+tanh((v-v3)/v4))
tauw(v)=1/cosh((v-v3)/(2*v4))
aux ica=gca*minf(v)*(v-vca)
param i=60,c=20
param vk=-84,vl=-60,vca=120
param gk=8,gl=2,gca=4.4
param v1=-1.2,v2=18,v3=2,v4=30,phi=.04
init v=-60,w=0.015
#@range i 0 300
@ total=400,dt=0.25,meth=rk4,xp=v,yp=w
@ xlo=-75,xhi=45,ylo=-0.1,yhi=0.55
done
```

Table 2 summarises the language. The `#@range` line is an XPP Studio extension that sets a slider's range. Because it starts with `#`, XPPAUT reads it as a comment, so files stay portable.

Methods. `meth=` accepts `rk4` (fixed-step Runge–Kutta), `euler`, `dp45` (adaptive Dormand–Prince 4(5)), and the stiff names `stiff`, `gear`, `cvode`, `rosenbrock` and `backeul`. All of the stiff names select an adaptive Rosenbrock 2(3) solver [8]. `discrete` iterates a map. Models with Wiener variables always use Euler–Maruyama, and models with Volterra integrals use a product-trapezoid scheme.

Syntax	Meaning
`x'=...  or  dx/dt=...`	differential equation
`x(t+1)=...`	map (or use `x'=...` with `@ meth=discrete`)
`x(0)=1  or  init x=1,y=0`	initial values
`par a=1, b=2`	parameters (also `param`, `p`)
`number c=3`	named constant
`!d=a*b`	derived parameter, recomputed when parameters change
`f(x,y)=...`	user function
`q=...`	fixed quantity, evaluated before the equations
`aux s=...`	auxiliary output, plotted like a variable
`wiener w`	white noise for an SDE, e.g. `x'=-x+s*w`
`markov z 2  {0} {kon}  {koff} {0}`	two-state Markov variable; entry (i, j) is the rate from i to j
`table h % 101 0 6.283 sin(t)`	table sampled from an expression
`table h data.tab`	table read from a file (add it with **Table files**)
`b u-1  b u'-2`	boundary conditions: `u` at the left end, `u'` at the right end (also `bndry`)
`#@irange u lo hi`	slider range for the initial value of `u`
`x[1..20]'=-x[j]+x[j-1]`	array: the line is repeated for $j = 1, \ldots, 20$
`%[1..20] ... %`	repeat a block of lines
`int{K(t,t',x)}`	Volterra integral $\int_0^t K(t, t', x(t'))\, dt'$
`int{K#x}`	convolution $\int_0^t K(t-t')\, x(t')\, dt'$
`int[mu]{...}`	weakly singular kernel, extra factor $(t-t')^{-\mu}$
`u(t)=...int{...}`	integral equation (also `volterra u=...`)
`delay(x,tau)`	delayed value $x(t-\tau)$; τ may depend on the state
`x(0)=0.5+0.2*cos(t)`	history of a delay equation for $t < 0$ (its value at $t = 0$ is the initial condition)
`global 1 x-1 {x=0;y=y+1}`	global flag: when `x-1` crosses zero upwards (1), downwards (-1) or either way (0), make the assignments; `arret=1` stops the run
`#@range p lo hi`	slider range for parameter `p`
`@ total, t0, trans, dt, meth, toler, bound`	integration (`t0` is the start time): duration, transient, step, method, tolerance, bound
`@ xp, yp, zp, axes=3, xlo, xhi, ylo, yhi`	initial plot variables and window
`@ fold=th, tor_per=6.2832`	periodic variable `th` with the given period
`@ seed=7`	fix the random seed
`@ vmaxpts=4000`	memory length for Volterra integrals
`@ ds, dsmin, dsmax, nmax, ntst, parmin, parmax`	continuation numerics, as in XPPAUT's AUTO window
Functions: `sin cos tan exp ln log sqrt abs heav sign max min mod atan2 sinh cosh tanh`, `if(c)then(a)else(b)`, ...	

Table 2: The `.ode` language understood by XPP Studio.

4 Tutorial 1: phase-plane analysis

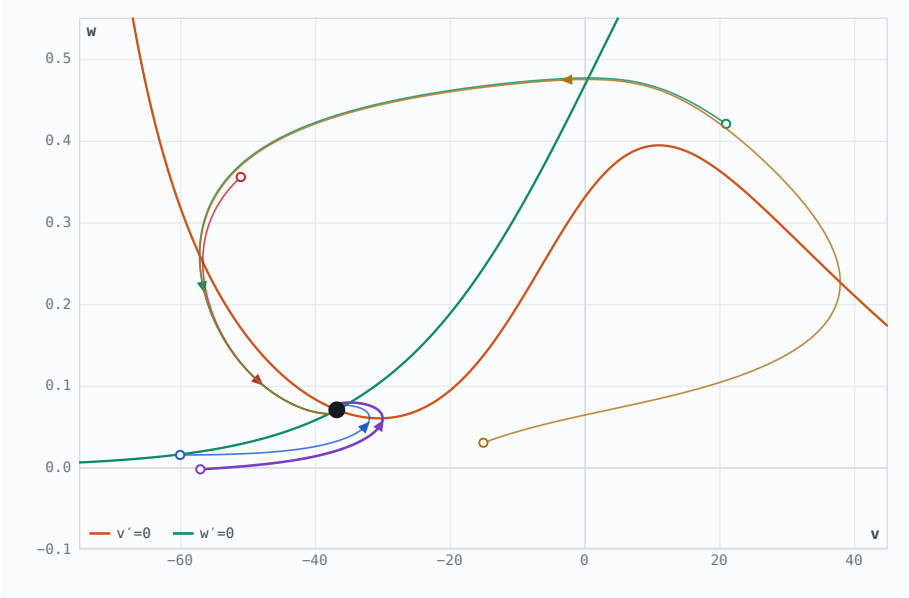


Figure 2: Morris–Lecar phase plane at $i = 60$: the v-nullcline (orange), the w-nullcline (green), a stable focus (filled circle) and several trajectories. Open circles mark starting points.

Choose **Morris–Lecar** from the **Model** menu. The phase plane shows w against the membrane potential v, together with the two nullclines.

Try it

1. Click anywhere in the plane to start a trajectory there. Shift-click integrates in both directions of time. Each new trajectory gets its own colour and appears in the **Trajectories** list, where you can remove it with ×.
2. Press D for the direction field and N to hide or show the nullclines. Hovering over the plot shows the coordinates and the derivatives v', w' at the pointer.
3. The inspector lists one equilibrium, a *stable focus*, with its eigenvalues $\lambda \approx -0.055 \pm 0.063i$. **Set as IC** moves the initial condition onto it.
4. Open the **Parameters** tab and slowly drag the slider for the applied current i. Between about $i = 94$ and $i = 212$ the focus becomes unstable and trajectories settle onto a large limit cycle: repetitive firing.
5. Press 3 for the time course. The chips in the inspector choose which variables are plotted, auxiliary ones included (`ica` here).
6. To label the figure, press **Note**, click next to the stable focus and type `stable focus`. Drag the label where you want it, and choose a colour in the box if you like. Labels are explained in Section 2.

Equilibria are found by Newton's method from a grid of starting points covering the visible window, so panning or zooming out searches a wider region. For a saddle, the **Separatrices** button integrates its stable and unstable manifolds. Once drawn, they follow the saddle: when you change a parameter, the saddle is found again and its separatrices are recomputed from its new position, so you can watch, for example, a separatrix approach a homoclinic connection as you drag a slider.

Many trajectories at once: directions and grids

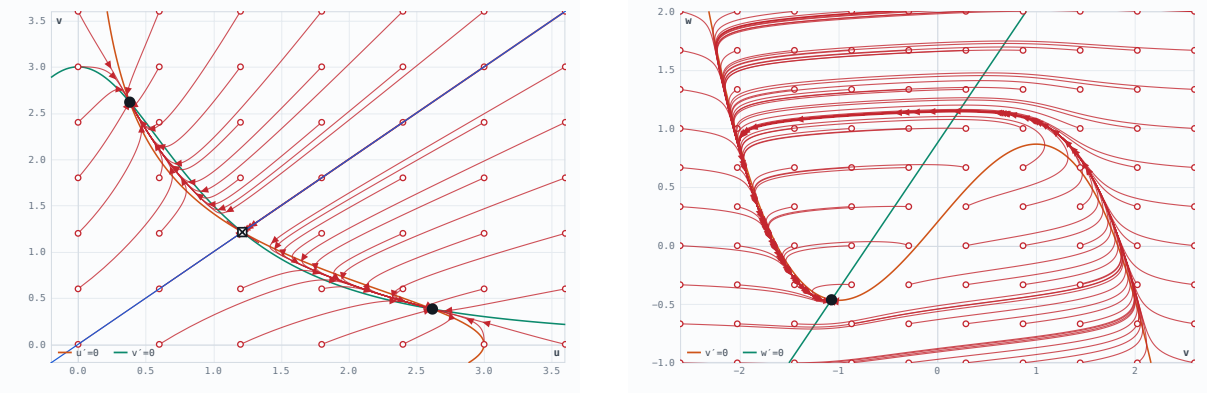


Figure 3: Left: the genetic toggle switch with a 7×7 grid of initial conditions from **Run grid**, u and v each running from 0 to 3.6, and the separatrices of the saddle. Every trajectory ends at one of the two stable nodes, and the stable manifold of the saddle (blue) separates the two basins. Right: **Run grid** with its default settings on the FitzHugh–Nagumo phase plane: trajectories from a 10×10 grid covering the window.

Direction. A run can go forward in time, backward, or in both directions. Choose this in the **Initial** tab, next to **Integrate**, or under **Direction** in the Numerics tab; the two stay in step. *Both* integrates backward and forward from the same point and joins the two halves into one trajectory, with t running from $-T$ to T, so the time course and the arrow on the phase plane show it as a single curve. Trajectories run backward are marked ← in the list and those run both ways ↔. Shift-clicking the plane always runs both ways. Maps, delay and Volterra equations run forward only.

Several initial conditions. XPPAUT's *Initialconds › Range* and *Flow* integrate from many starting values at once. In XPP Studio both are done by the **Several initial conditions** section of the **Initial** tab (Figure 4). Choose one or two variables, each with a start value, an end value and a number of steps, then press **Run grid**; every combination is run, in the chosen direction, while the other variables keep their values from the list above.

By default the grid is 10×10 over the phase-plane window, using the two plotted variables and their axis limits, which gives a quick picture of the whole phase portrait (Figure 3, right). These defaults follow the axes and window as you change them, until you edit a value yourself; **Use window** goes back to them. Set the second variable to *(none)* for a single row of starting points. The runs share one colour and appear in the trajectory list, **Stop** ends a long grid early, and E erases the result. Figure 3 (left) shows a 7×7 grid on the toggle switch.

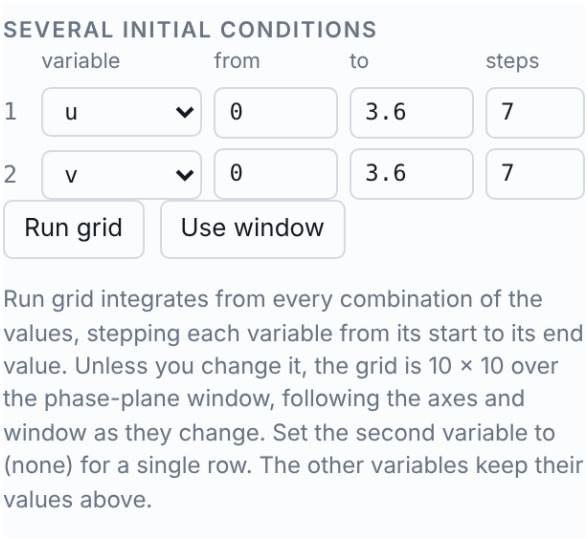


Figure 4: The **Several initial conditions** section of the Initial tab, set for a 7×7 grid in u and v.

5 Tutorial 2: one-parameter bifurcation diagrams

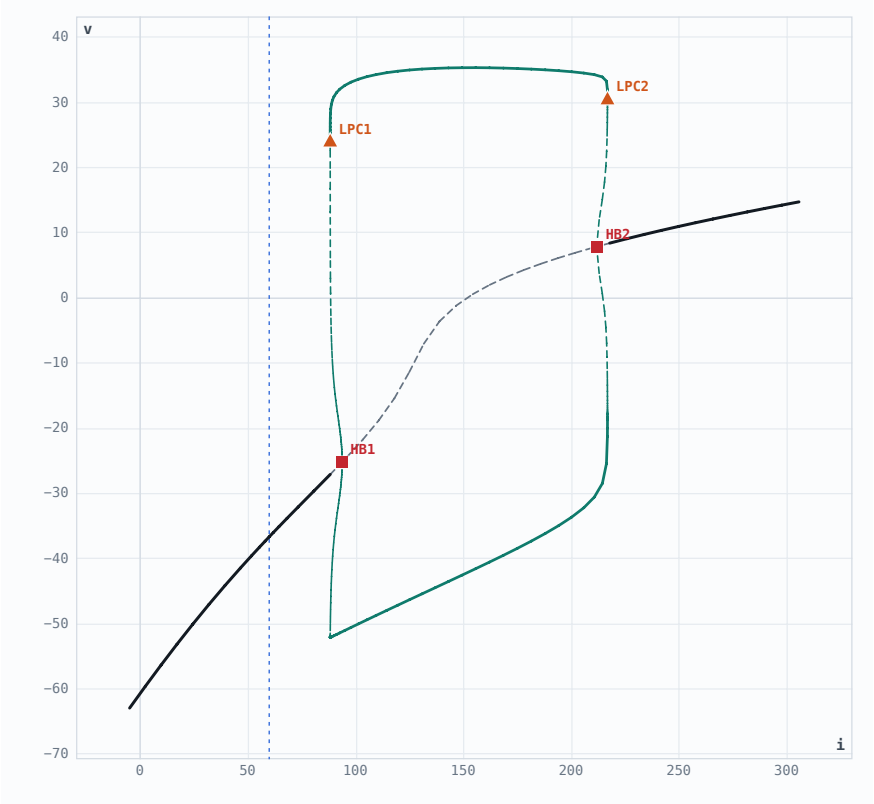


Figure 5: Bifurcation diagram of Morris–Lecar in the current i. Black: equilibria (solid stable, dashed unstable). Green: maximum and minimum of v along the periodic orbits born at the two Hopf points HB1 and HB2. Both Hopf bifurcations are subcritical. The orbit family turns at two folds of cycles, LPC1 and LPC2, giving bistability between rest and firing.

Press 4 for the Bifurcation view. The inspector shows the continuation controls: the **Parameter** to vary, the variable to **Plot**, the parameter **Range**, the initial step **Step ds**, and the equilibrium to **Start** from.

Try it

1. Press **Continue equilibria**. Pseudo-arclength continuation [7] traces the branch in both directions across the range. Stability is read off the eigenvalues, and two Hopf points appear: HB1 at $i \approx 93.68$ (period ≈ 79.2) and HB2 at $i \approx 212.06$. Each Hopf point is tagged *sub* or *super* in the **Special points** table (Figure 9). The tag comes from the sign of the first Lyapunov coefficient ℓ_1, and hovering over it shows the value. Both points here are subcritical ($\ell_1 > 0$), so the orbits born there are unstable.

2. In the **Special points** table, press **Cycles** on the row for HB1. The periodic orbits born there are computed by multiple shooting with an integral phase condition, as in AUTO, and their stability comes from Floquet multipliers. While the computation runs, Esc (or **Skip direction**) abandons the current direction and moves on, and Shift+Esc stops altogether.

3. The family of orbits connects HB1 to HB2 and turns at two folds of cycles: LPC1 at $i \approx 88.29$ and LPC2 at $i \approx 216.90$ (Figure 5). Between LPC1 and HB1 a stable orbit and the stable rest state coexist.

4. Click any point of a branch to jump there. This sets the parameter and loads that equilibrium, or exactly one period of that orbit, as the current state. Clicking a row of the special-points table does the same.

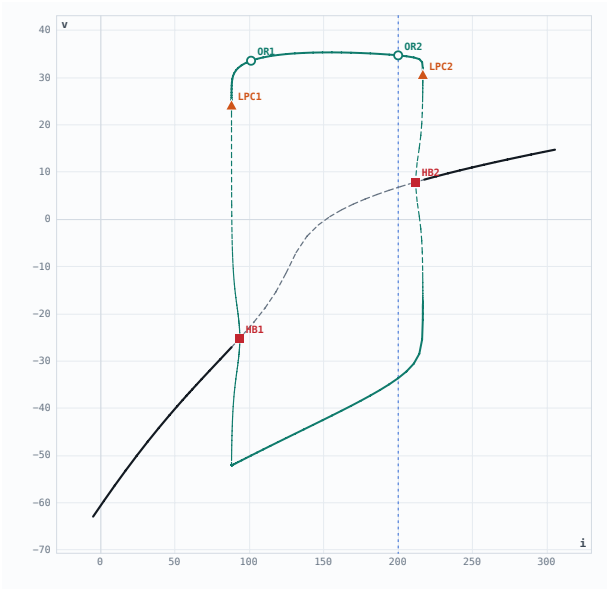
5. **Limit cycle** starts orbit continuation from the current trajectory instead. Integrate onto a stable cycle first (e.g. at $i = 150$), then press it.

6. **Show** sets what the vertical axis measures for the variable chosen under **Plot**. The default *hi-lo* draws the maximum and minimum of the variable over each orbit. *L2 norm* draws $\left(\frac{1}{T}\int_0^T x(t)^2\,dt\right)^{1/2}$ over each orbit, and $|x|$ at each equilibrium, so a branch of orbits becomes a single curve that leaves the equilibrium branch at the Hopf point (Figure 7). The norms are computed the first time you choose this, by integrating one period of each orbit. *Period* and *frequency* draw the period of the orbits, or its inverse, against the parameter (Figure 8). The period grows from about 79 at HB1 to 135 at LPC1 and is about 42 at HB2. Sweeps show their periods too, where the oscillation is regular enough for one to be estimated. Changing **Show** or **Plot** refits only the vertical axis, to the data in the range of the parameter on screen, so a part of the diagram you have zoomed into stays in view as you switch between hi-lo, L2 norm and period.

7. Click a point on the orbit branch. As well as loading that orbit, this adds it to the **Special points** table as OR1, marked on the diagram with an open circle (Figure 6). Click another point to add OR2, and so on. Each picked orbit has a **Const. T** button, which follows it in two parameters at fixed period (Tutorial 3), and a **Remove** button.

8. **Deleting single curves.** Clicking a curve also selects it: the branch, orbit family, sweep or boundary-value branch is outlined in a blue halo (Figure 7), and the status line names it. **Delete curve** under the continuation buttons, or the Delete key, removes that curve and its special points, while the rest of the diagram stays. Esc clears the selection. **Clear diagram** still removes everything for the current parameter. The Two-parameter view works the same way, with its own **Delete curve** button.

Sweep max/min provides a brute-force alternative. It integrates at a grid of parameter values, carrying the end state forward, and plots every local maximum and minimum after the transient. This is the right tool for chaotic or stochastic regimes, where continuation does not apply (Tutorial 4).



SPECIAL POINTS 6

PT	I	V	PERIOD	
HB1 sub	93.682	−25.321	79.25	Follow
				Cycles
HB2 sub	212.06	7.8003	42.33	Follow
				Cycles
LPC1	88.293	23.925	135.4	Follow
LPC2	216.9	30.397	77.93	Follow
OR1	101.44	33.496	83.99	Const. T
				Remove
OR2	200.31	34.634	65.68	Const. T
				Remove

Figure 6: Left: two periodic orbits picked on the Morris–Lecar orbit branch, OR1 at $i \approx 100$ ($T \approx 84.0$) and OR2 at $i \approx 200$ ($T \approx 65.7$). Right: the **Special points** table with the picked orbits at the bottom, each with **Const. T** and **Remove**.

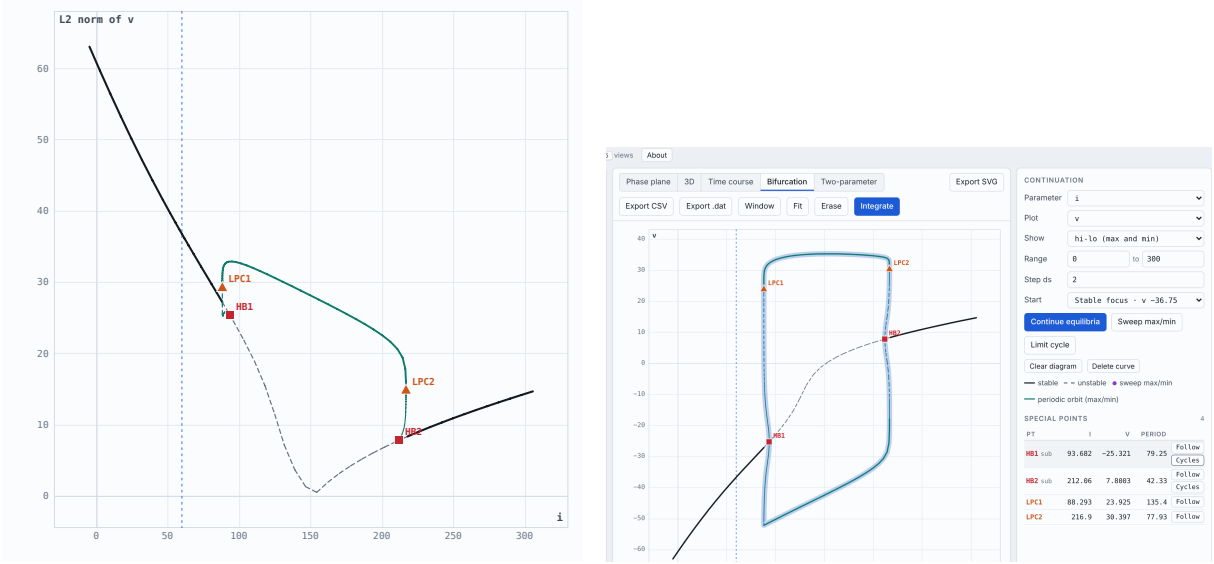


Figure 7: Left: the Morris–Lecar diagram with **Show** set to *L2 norm* of v. The orbit family is now one curve from HB1, through the folds LPC1 and LPC2, to HB2. Right: the orbit family selected by clicking it, outlined in blue, with **Delete curve** enabled under the continuation buttons.

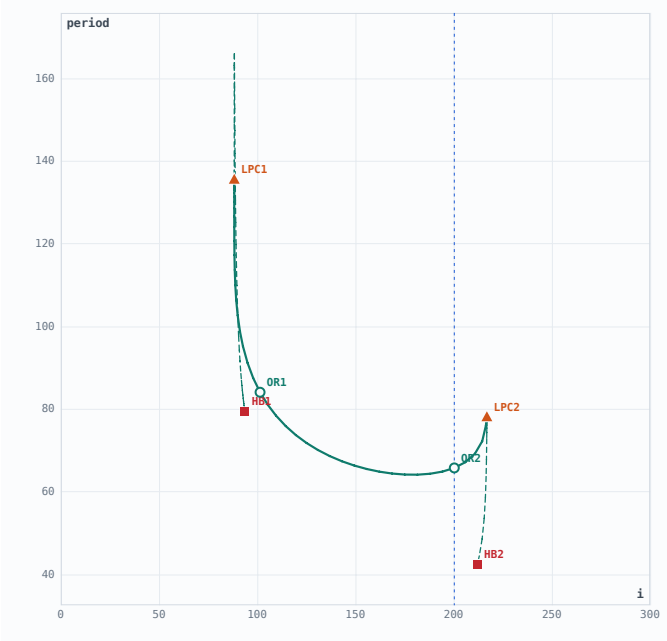


Figure 8: The same diagram with **Plot** set to **period**: the period of the periodic orbits against i, with the special points and the picked orbits.

SPECIAL POINTS 2

PT	I	V	PERIOD	
HB1 sub	93.682	−25.321	79.25	Follow Cycles
HB2 sub	212.06	7.8003	42.33	Follow Cycles

Figure 9: The **Special points** table after continuing equilibria of Morris–Lecar. Both Hopf points are tagged *sub* (subcritical). **Follow** continues a point in two parameters, and **Cycles** continues the periodic orbits born there.

Sub- or supercritical? At a Hopf point with eigenvalues $\pm i\omega$, XPP Studio computes the first Lyapunov coefficient

$$\ell_1 = \frac{1}{2\omega}\,\mathrm{Re}\Big[\langle p, C(q,q,\bar{q})\rangle - 2\langle p, B(q, A^{-1}B(q,\bar{q}))\rangle + \langle p, B(\bar{q}, (2i\omega I - A)^{-1}B(q,q))\rangle\Big],$$

where A is the Jacobian, $Aq = i\omega q$, $A^{\mathsf{T}}p = -i\omega p$ and $\langle p, q\rangle = 1$ [6]. The multilinear forms B and C (the second and third derivatives of the vector field) are evaluated by finite differences. If $\ell_1 < 0$ the bifurcation is *supercritical*: a small stable orbit appears on the side where the equilibrium is unstable. If $\ell_1 > 0$ it is *subcritical*: an unstable orbit appears on the side where the equilibrium is stable, and the system usually jumps to a distant attractor.

What is detected. For equilibria of flows: folds (LP), Hopf points (HB, tagged sub- or supercritical) and branch points (BP). For periodic orbits: folds of cycles (LPC), period doublings (PD) and torus (Neimark–Sacker) points (TR). For fixed points of maps: LP, PD

and Neimark–Sacker (NS). At a PD of cycles, **Double** switches to the period-doubled branch, and at a branch point **Switch** follows the other branch through it. A branch that comes back to its starting point, such as an isola, is closed into a loop and reported as such.

6 Tutorial 3: two-parameter continuation



Figure 10: Left: two-parameter diagram for Morris–Lecar in (i, ϕ). A single Hopf curve joins HB1 and HB2. It is purple where the Hopf bifurcation is subcritical and red where it is supercritical, and the colour changes at the generalised Hopf (Bautin) points GH1 and GH2. Dashed ends are neutral saddles beyond the Bogdanov–Takens points BT1 and BT2 near $\phi = 0$. The green fold-of-cycles curve from LPC1 ends at GH3, which lies next to GH1. The blue cross-hairs mark the current parameter values. Right: the Two-parameter panel, with a colour picker for each kind of curve and the list of codimension-two points.

Any special point in the table with a **Follow** button can be continued in a second parameter. The Two-parameter view (5) shows the result, and its inspector panel selects the two parameters and their ranges.

Try it

1. After Tutorial 2, choose the second parameter (here ϕ, the rate of the recovery variable) in the Two-parameter panel.
2. Set the range of ϕ to 0–0.4. Then press **Follow** on HB1 (in the Bifurcation view, or choose HB1 under **Start** and press **Follow curve**). Hopf curves are computed from a

bordered system with the bialternate product [5, 6]. Here a single closed arch joins HB1 to HB2, so following HB2 as well would only retrace it.

3. The first Lyapunov coefficient ℓ_1 is computed at every point of the Hopf curve, and the curve is coloured by its sign: purple for subcritical, red for supercritical (Figure 10). Where ℓ_1 changes sign there is a *generalised Hopf* or *Bautin* point, labelled GH. For Morris–Lecar these are at about $(i, \phi) = (124.1, 0.300)$ and $(165.9, 0.253)$, so the Hopf bifurcations are supercritical only near the top of the arch. Bogdanov–Takens (BT), cusp (CP) and zero-Hopf (ZH) points are flagged in the same way. For maps, strong resonances R1 and R2 and fold–flip points are flagged too.

4. Press **Follow** on LPC1 to continue the fold of cycles. Its curve is computed by multiple shooting and is slower, typically 10–20 seconds. Bautin theory predicts that a curve of folds of cycles is born at every GH point, and here the LPC curve runs up the left side of the arch and ends at GH3 $\approx (124.4, 0.306)$. This is the endpoint of the orbit computation and agrees closely with GH1 from ℓ_1, an independent check of both.

5. Each kind of curve has a colour swatch in the panel legend: fold, sub- and supercritical Hopf, neutral saddle, fold of cycles, constant period and period doubling. Click a swatch to choose a new colour. The choice applies at once, is used by **Export SVG**, and is remembered in your browser. **Default colours** goes back to the theme colours.

6. Click a point on any curve to move to those parameter values. Then return to the phase plane or the bifurcation diagram to see the dynamics there.

7. **Curves of constant period.** Pick one or more orbits on the bifurcation diagram (Tutorial 2) and press **Const. T** on one of them, or choose it under **Start** as "OR1 (T=...)" and press **Follow curve**. The orbit is continued in both parameters with its period held fixed: the unknowns are the orbit, i and ϕ, and the extra equation is $T = T_0$. In Figure 11 the two gold curves are the parameter values where Morris–Lecar oscillates with period 84.0 and with period 65.7. Each ends where the orbit shrinks onto an equilibrium at a Hopf point whose period $2\pi/\omega$ equals T_0, so constant-period curves end on the Hopf curve. Integrating at points along a curve gives the same period to within the output spacing.

Curves of period doublings of cycles (PDC) are continued the same way, which gives the classical period-doubling tongues of the Rössler system.

Legends, slices and a third parameter.

- **Legend**, above the Bifurcation and Two-parameter plots, adds a key with one line per curve: what it is, where it was started from, and any parameter values that differ between curves. The legend is included in **Export SVG**. It starts in the top right corner of the plot; drag it elsewhere, and double-click it to send it back. **Colours**, next to **Legend**, gives every curve its own colour instead of colouring by kind, so that the legend lines can be told apart.
- **Slices in a third parameter**, at the bottom of the Two-parameter panel, repeat the curve chosen under **Start** at several values of a third parameter. Choose **Parameter 3**, its smallest and largest values and the number of slices, and press **Compute slices**. Each slice starts from the special point on the previous one, so the curves are followed continuously as the third parameter changes. This works for fold and Hopf curves, and for folds of cycles that start from a Hopf point of the diagram.
- **3D**, above the Two-parameter plot, draws the curves in a box with the third parameter as the vertical axis, each curve at the value of that parameter it was computed with (Figure 12). Drag to rotate and scroll to zoom, as in the 3D view of trajectories.

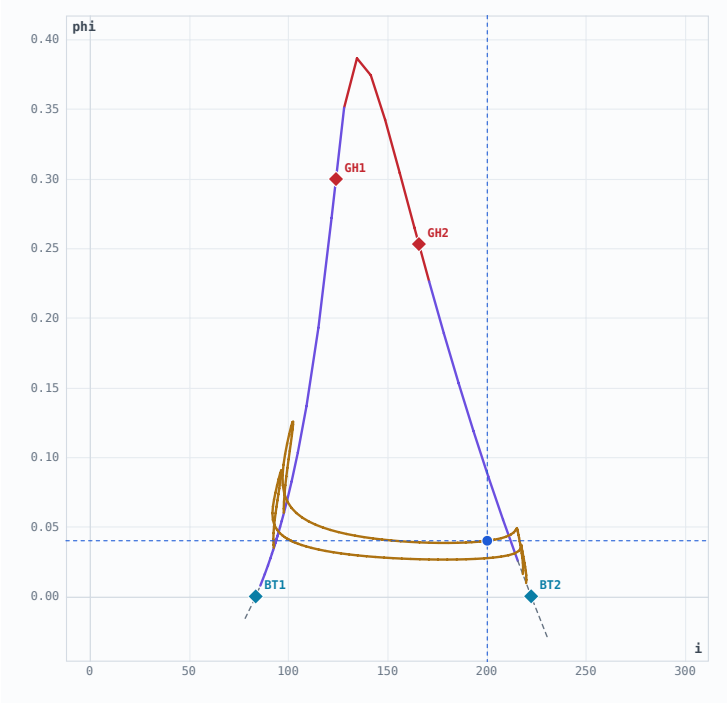


Figure 11: Curves of constant period (gold) for Morris–Lecar in (i, ϕ), traced from the picked orbits OR1 ($T \approx 84.0$) and OR2 ($T \approx 65.7$), together with the Hopf curve, coloured by criticality. Each constant-period curve ends on the Hopf curve, where the frequency of the Hopf point matches the fixed period.

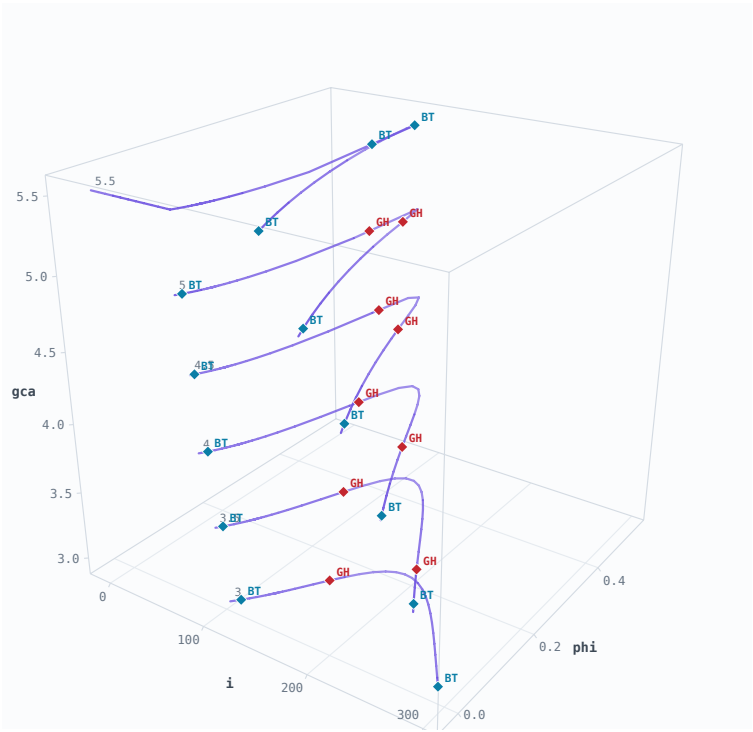


Figure 12: Slices: the Hopf curve of Morris–Lecar in (i, ϕ) followed at six values of g_{Ca} from 3 to 5.5 and drawn in 3D, each curve at the height of its value of g_{Ca}. The generalised Hopf points (GH) and Bogdanov–Takens points (BT) on each slice trace out how they move with the third parameter.

Try it

1. After following HB1 above, set **Parameter 3** to g_{Ca} with values 3 to 5.5 and 6 slices, and press **Compute slices**.
2. Turn on **3D** and rotate the box. As g_{Ca} decreases the arch of Hopf points shrinks, and the two GH points move together.

7 Tutorial 4: chaos in three dimensions

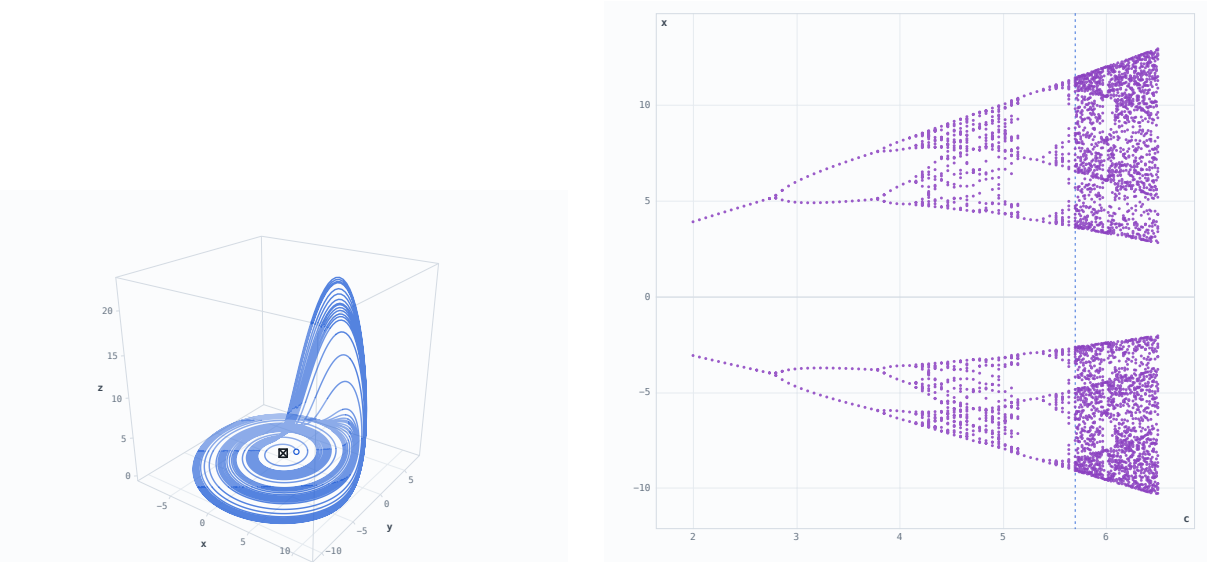


Figure 13: Rössler system at $a = b = 0.2$. Left: the chaotic attractor at $c = 5.7$ in the 3D view. Right: **Sweep max/min** in c, showing the local maxima and minima of x and the period-doubling route to chaos.

Load **Rössler system** [15] and press 2. Drag to rotate the attractor, scroll to zoom, or press **Spin**. The axes are chosen in the toolbar, and any of them can be time.

Try it

1. In the Bifurcation view choose c and press **Sweep max/min** (Figure 13, right). The number of parameter values is set by **Sweep samples** in the Numerics tab.
2. For a continuation version, set $c = 2.5$, integrate onto the stable cycle and press **Limit cycle**. Period doublings are found at $c \approx 2.8324$, 3.8374, 4.1242 and 4.1870. Pressing **Double** at each in turn follows the cascade.
3. **Follow** on a PD point traces its curve in (c, a).

8 Tutorial 5: maps

A model is a map when its equations are written `x(t+1)=...`. Trajectories are then drawn as points, and for a one-variable map the phase plane becomes a cobweb diagram (Figure 14). For maps, **Sweep max/min** keeps up to 250 iterates after the transient, which gives the familiar orbit diagram. Continuation finds folds, period doublings and Neimark–Sacker points of fixed points. The **Hénon map** example shows a fold at $a \approx -0.1225$ and a PD at $a \approx 0.3675$, and the **Delayed logistic map** an NS bifurcation at $a = 2$.

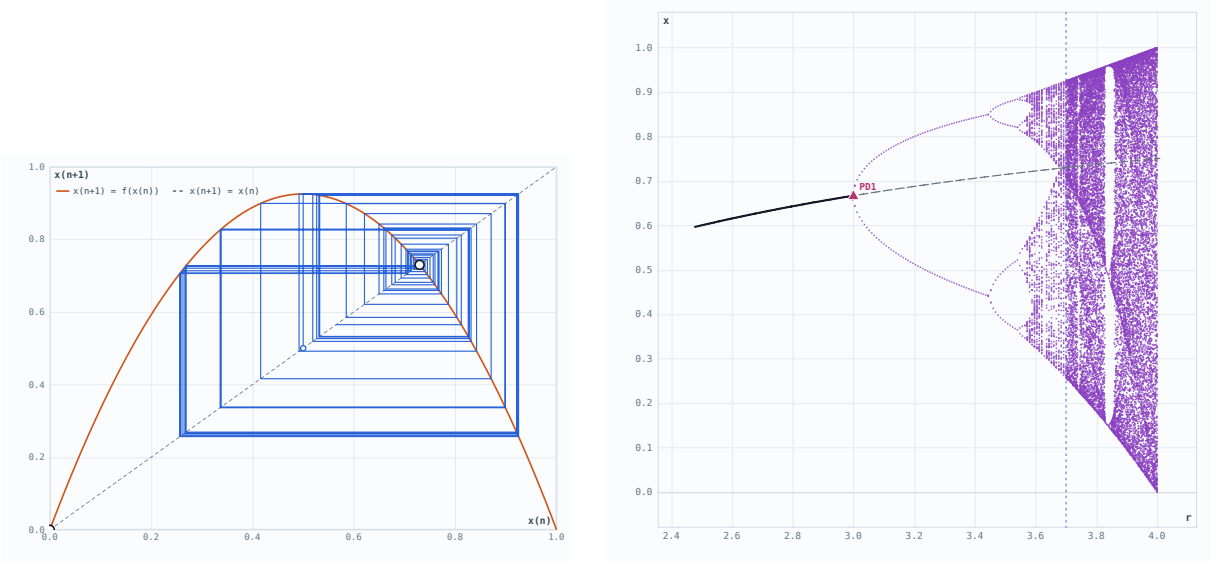


Figure 14: Logistic map $x \mapsto rx(1-x)$. Left: cobweb diagram at $r = 3.7$. Right: orbit diagram from **Sweep max/min** (purple dots), with the continued fixed point and its period doubling PD1 at $r = 3$.

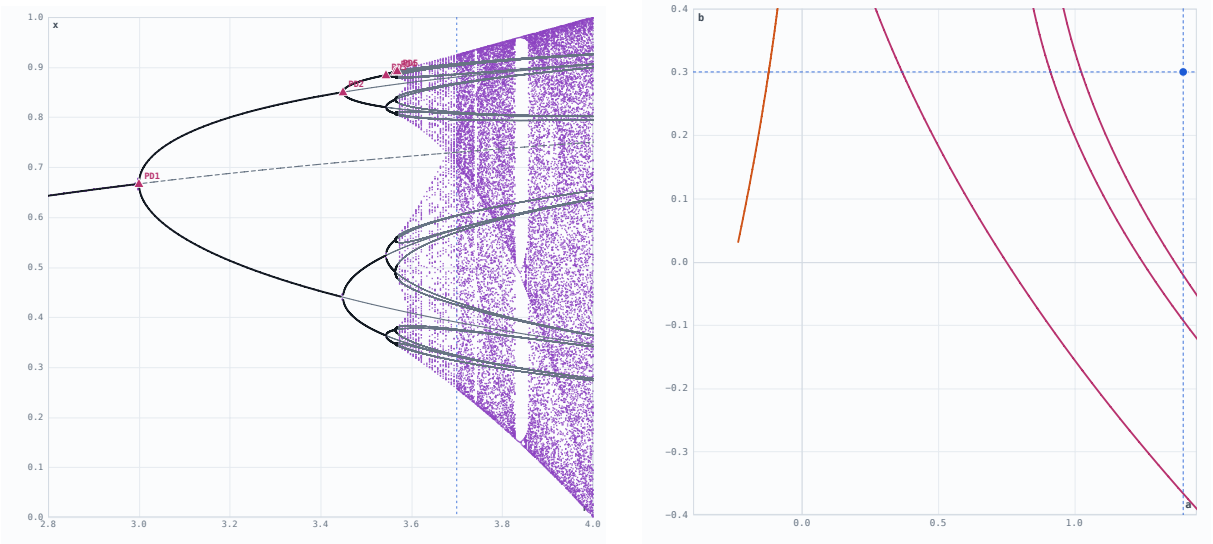


Figure 15: Following period doublings of maps. Left: the logistic map, with the fixed point and the orbits of period 2, 4, 8 and 16 continued by pressing **Double** at PD1 to PD4 in turn, over the orbit diagram. Each orbit is drawn at all of its points; stable parts are solid and black, and each orbit carries on, unstable (grey, dashed), from its next period doubling to $r = 4$. Right: the Hénon map in (a, b), with the fold curve of the fixed point (orange) and the period-doubling curves (crimson) of the fixed point and of the orbits of period 2 and 4, from left to right.

Period doublings and orbits of higher period

An orbit of period N of a map f is a fixed point of the N-th iterate f^N, and XPP Studio continues it as one. Its stability comes from the multipliers of f^N, and a period doubling of f^N is where the orbit of period $2N$ is born.

Try it

1. Load **Logistic map**, open the Bifurcation view and press **Continue fixed points**. The fixed point $x = 1 - 1/r$ loses stability at PD1, $r = 3$.
2. Press **Double** on PD1 in the table of special points. The orbits of period 2 are continued, and each is drawn at both of its points, so the branch splits into two. A new period doubling PD2 is found at $r \approx 3.4495$, and the **Period** column of the table gives the period of the orbit it creates, here 4.
3. Press **Double** on PD2, then on PD3 and PD4. The doublings come at $r \approx 3.5441$ and 3.5644, closer and closer together, with ratios of successive gaps approaching Feigenbaum's constant 4.669… [4]. Press **Orbit diagram** to compare with the attractor (Figure 15, left). Where a continued orbit is drawn dotted, it is momentarily an orbit of lower period.
4. Orbits that are not born in a period doubling can be followed directly. Set **Period** to 3 in the Continuation panel, set $r = 3.84$ and $x = 0.15$, and press **Continue fixed points**. The period-3 window opens at a fold, LP1 at $r = 1 + \sqrt{8} \approx 3.8284$, and its stable orbit doubles at $r \approx 3.8415$.
5. **Follow** on a period doubling traces its curve in two parameters, for any period. In the **Hénon map** example, continue the fixed point in a, press **Double** on PD1 and PD2, set the second parameter to b, and follow PD1, PD2 and PD3 (Figure 15, right). For the fixed point the curve is $a = \frac{3}{4}(1-b)^2$.

9 Tutorial 6: noise and random switching

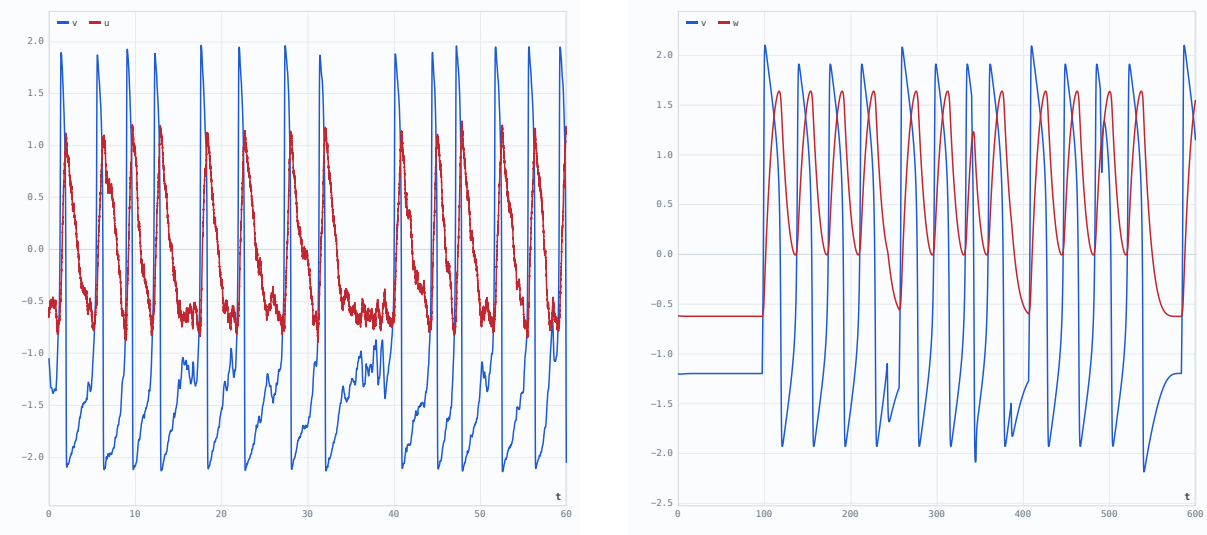


Figure 16: Left: excitable FitzHugh–Nagumo driven by white noise; noise triggers irregular spikes. Right: FitzHugh–Nagumo whose input is a two-state Markov variable, so oscillation switches on and off at random.

Wiener noise. Declaring `wiener w` makes `w` a white-noise source. The equations are integrated by Euler–Maruyama with step `dt`, and over each step `w` is $N(0,1)/\sqrt{dt}$, exactly as in XPPAUT. In **Noisy FitzHugh–Nagumo** the rest state is stable, and noise of amplitude `sig` kicks the system over threshold (Figure 16, left). Drag `sig` to change the firing rate.

Markov variables. **Telegraph-driven FitzHugh–Nagumo** declares

```
markov s 2
{0} {kon}
{koff} {0}
```

a variable that jumps $0 \to 1$ at rate `kon` and $1 \to 0$ at rate `koff`. Rates may depend on the state variables. The input `i0+di*s` switches the cell between rest and oscillation (Figure 16, right).

Seeds. Each stochastic run uses a new random seed, which is reported in the status line. Enter a number in **Noise seed** (Numerics tab), or write `@ seed=7`, to repeat the same noise path, for instance to compare parameter values. Equilibria, nullclines and continuation always use the noise-free system, with Markov states held at the values in the Initial tab.

10 Tutorial 7: stiff systems

The **Stiff van der Pol** example has $\mu = 1000$: relaxation oscillations whose slow and fast time scales differ by a factor of about 10^6. With `@ meth=stiff` the Rosenbrock 2(3) solver integrates a period ($T \approx 1614.4$) in a few thousand steps, where an explicit method needs hundreds of thousands. When the stiff method is selected, periodic-orbit continuation also uses it for its shooting segments, so the period can be continued down in μ.

11 Tutorial 8: arrays and Volterra equations

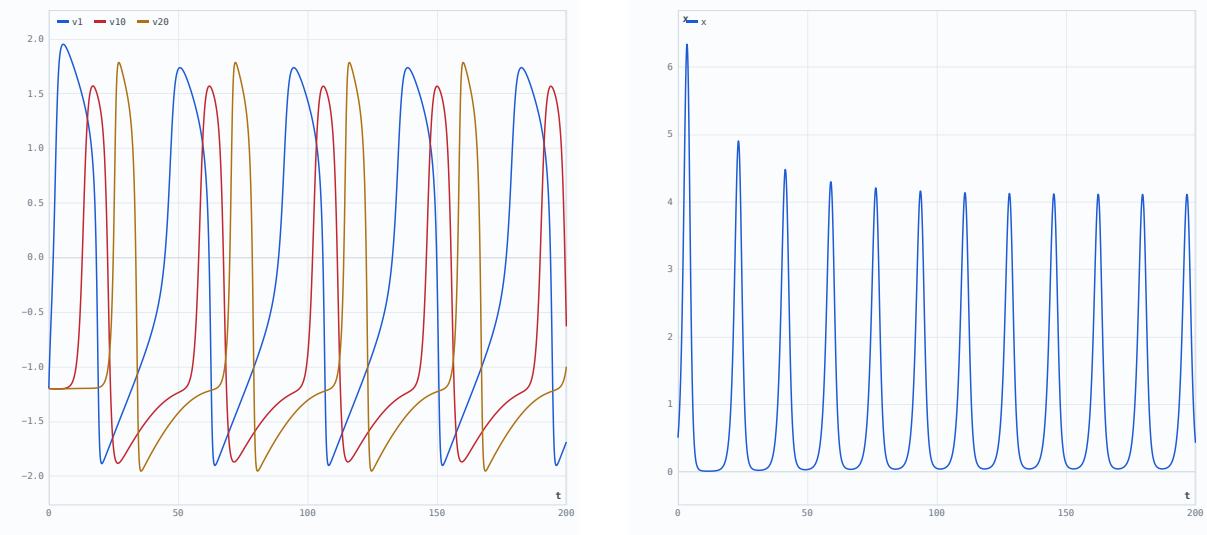


Figure 17: Left: a chain of 20 FitzHugh–Nagumo cells written with arrays. Spikes start at cell 1 and arrive later at cells 10 and 20. Right: logistic growth with a gamma-distributed delay, written as a Volterra convolution. At $r\tau = 2.5$ the equilibrium is unstable and the population oscillates.

Arrays. XPPAUT's arrays are a text macro. The **Chain of 20 FitzHugh–Nagumo cells** example is written as

```
%[1..20]
v[j]'=v[j]-v[j]^3/3-w[j]+i1*heav(1.5-[j])+d*(v[j+1]-2*v[j]+v[j-1])
%
w[1..20]'=eps*(v[j]+a-b*w[j])
v0=v1
v21=v20
init v[1..20]=-1.2
init w[1..20]=-0.62
```

The block between `%[1..20]` and `%` is repeated for $j = 1, \ldots, 20$. Every `[expr]` is replaced by its integer value, so `v[j-1]` becomes `v3` when $j = 4$, and `[j]` on its own becomes the number j. A single line containing `[1..20]` is repeated the same way. The load message reports how many lines the file expanded to. The expanded model is an ordinary 40-dimensional system, so its equilibria, eigenvalues (computed by the QR algorithm) and continuation all work as usual. In Figure 17 (left) the time course shows v_1, v_{10} and v_{20}; choose them with the chips in the inspector.

Space-time plots. For a chain or lattice it is often clearer to see all the cells at once, as in XPPAUT's array plot. In the Time course view, **Space-time** draws a row of indexed variables as an image, with time across and the cell index up (Figure 18). Each point is coloured by its value, and a colour bar shows the range. A *row* is any set of at least six variables with a common name and consecutive numbers, such as $v_1, \ldots, v_{20}$; when a model has several rows, **show** chooses between them. **colour min** and **max** fix the colour range, and **Auto** returns to the range of the run. In the chain the travelling pulses appear as stripes, and their slope is the propagation speed.

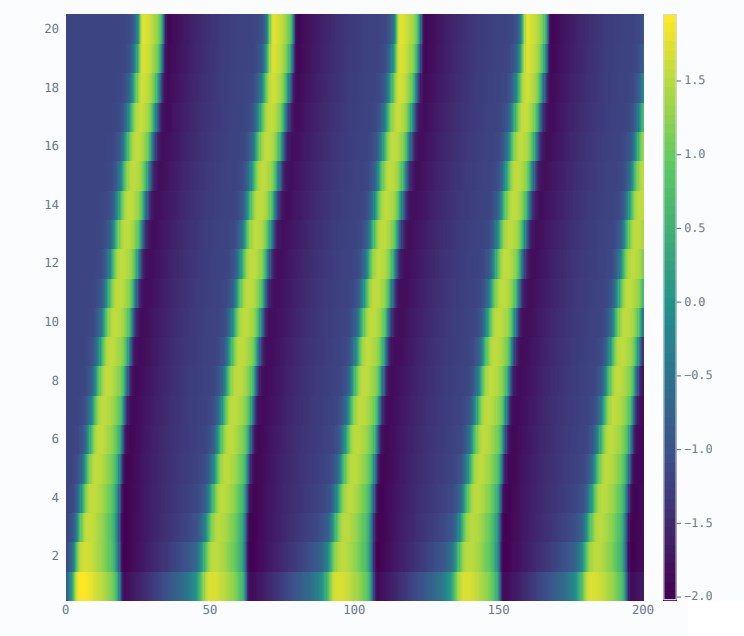


Figure 18: Space-time plot of $v_1, \ldots, v_{20}$ for the chain of FitzHugh–Nagumo cells. Each pulse starts at cell 1 and crosses the chain in about 20 time units.

Spectra. **Spectrum** in the Time course view replaces the time series by the power spectrum of each chosen variable, as amplitude against frequency (cycles per unit of t). A sinusoid of amplitude A gives a peak of height about A. **Log** uses a logarithmic amplitude axis, and **Smooth**

averages the spectra of overlapping segments of the run (Welch's method), which helps with noisy or chaotic runs at the cost of coarser frequency resolution.

Volterra integrals. XPP Studio supports XPPAUT's integral terms:

- `int{K(t,t',x)}` $= \int_0^t K(t,t',x(t'))\,dt'$, where the variables inside are evaluated at t';
- `int{K#x}` $= \int_0^t K(t-t')\,x(t')\,dt'$, a convolution with the kernel written as a function of t;
- `int[mu]{...}` adds the weakly singular factor $(t-t')^{-\mu}$, $0 \le \mu < 1$;
- `u(t)=...` or `volterra u=...` defines a variable by an integral equation.

The **Logistic growth with distributed delay** example is

```
x'=r*x*(1-int{t*exp(-t/tau)/tau^2#x})
par r=1,tau=2.5
@ total=200,dt=0.05,vmaxpts=1000
```

The kernel $K(s) = s\,e^{-s/\tau}/\tau^2$ has mean delay 2τ. Linearising about $x = 1$ gives $\lambda(1+\lambda\tau)^2 + r = 0$, so the equilibrium loses stability in a Hopf bifurcation at $r\tau = 2$. Sweeping τ with **Sweep max/min** shows the oscillation appear.

Integro-differential equations are solved by a fixed-step method: the trapezoid rule for the differential part and product-trapezoid quadrature [9] for the integrals. The quadrature weights are exact for the $(t-t')^{-\mu}$ factor, and the implicit terms are found by fixed-point iteration. The error falls as dt^2, or as $dt^{2-\mu}$ with a singular kernel. The method reproduces $u(t) = e^t$ for $u = 1 + \int_0^t u$, and the solution $e^{\pi t}\,\mathrm{erfc}(\sqrt{\pi t})$ of the Abel equation `u(t)=1-int[.5]{1#u}`. It also agrees with the equivalent three-variable ODE for the gamma kernel. Only the last `vmaxpts` steps are remembered (default 4000). Because the state of such a system is its whole history, equilibria, nullclines and continuation are switched off for these models. Trajectories and sweeps work normally.

Periodic variables and tables. Two further XPPAUT features are worth knowing:

- `@ fold=th` keeps an angle in $[0, 2\pi)$ (or $[0,$ `tor_per`$)$). Plots break the trajectory where it wraps, and continuation measures distances modulo the period; see the **Driven pendulum** example.
- Tables (**Phase oscillators with a tabulated H**) are interpolated linearly. They can be sampled from an expression or read from a data file.

12 Tutorial 9: delay equations

A delay-differential equation uses past values of its variables. In XPP Studio, as in XPPAUT, `delay(x,tau)` stands for $x(t-\tau)$, where `x` must be a variable with an equation and `tau` is any expression; it may depend on the state. Load **Delayed logistic equation**, Hutchinson's model [13]:

```
x'=r*x*(1-delay(x,tau))
aux xlag=delay(x,tau)
par r=1.8,tau=1
x(0)=0.5+0.2*cos(3*t)
```

History. A delay equation needs its state on a whole interval $[-\tau, 0]$, not only at $t = 0$. Writing `x(0)` as a formula in `t` gives that history, and its value at $t = 0$ becomes the initial condition. Without a formula, the history is held constant at the initial value. The formulas also appear in the **Initial** tab under **History for t < 0** (Figure 20), where they can be changed or cleared.

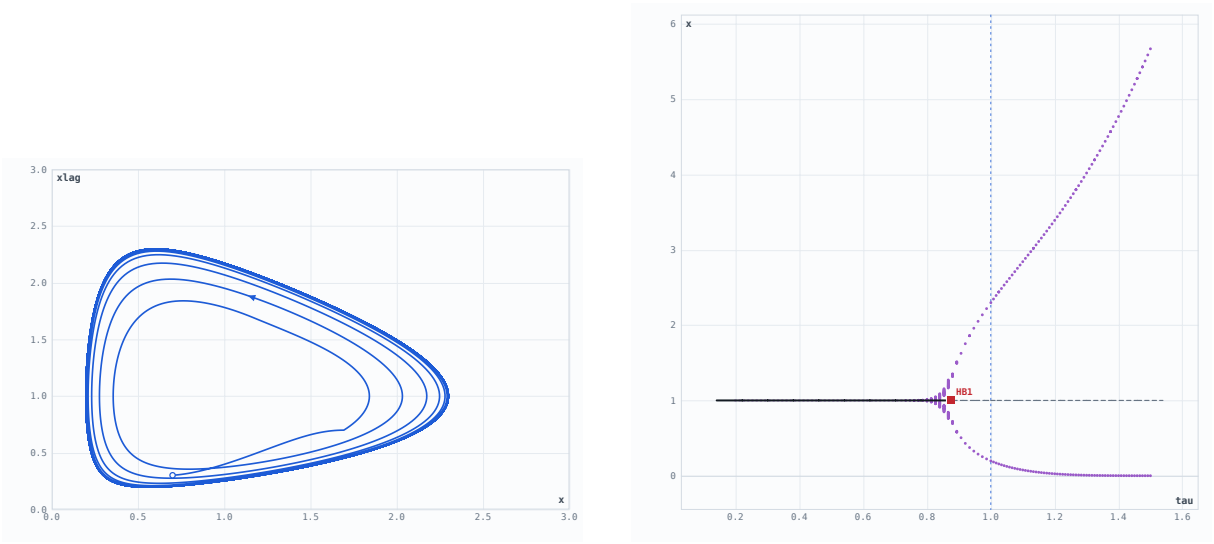


Figure 19: Hutchinson's delayed logistic equation with $r = 1.8$. Left: $x(t-\tau)$ against $x(t)$ for $\tau = 1$, starting from the history $0.5 + 0.2\cos 3t$ and settling onto a periodic orbit. Right: bifurcation diagram in τ. The equilibrium $x = 1$ (black) loses stability at the Hopf point HB1, $\tau \approx 0.874$, close to the exact value $\pi/(2r) = 0.873$. The purple dots, from **Sweep max/min**, show the oscillation that grows from it.

Clicking the phase plane changes only the values at $t = 0$. The auxiliary variable `xlag` makes $x(t-\tau)$ available for plotting, so the phase plane in Figure 19 shows $x(t-\tau)$ against $x(t)$, a common way to draw the attractor of a scalar delay equation.

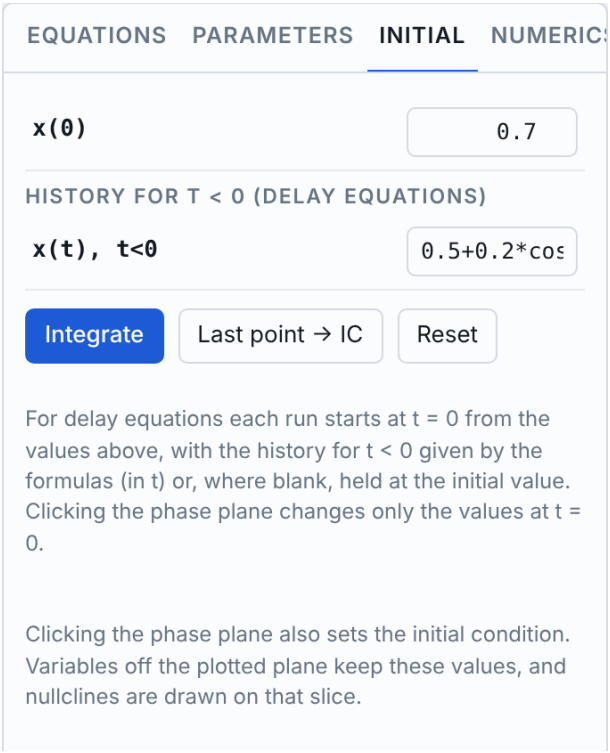


Figure 20: The Initial tab for a delay equation: the value at $t = 0$ and, below it, the history for $t < 0$ as a formula in t.

Integration. Delay equations run forward from $t = 0$ with fixed-step fourth-order Runge–Kutta (adaptive methods fall back to it). Every step is stored, and delayed values are found

by cubic Hermite interpolation of the stored solution, or from the history when $t - \tau < 0$. For $x' = -x(t-1)$ with $x = 1$ on $[-1, 0]$, whose exact solution is piecewise polynomial, the computed $x(2)$ and $x(3)$ agree with the exact values to rounding error.

Stability and continuation. Equilibria of a delay equation are those of the system with the delays removed, but their stability is not. It is set by the roots of the characteristic equation

$$\det\Big(\lambda I - A_0 - \sum_k B_k\, e^{-\lambda \tau_k}\Big) = 0,$$

where A_0 and B_k are the derivatives with respect to the current and the delayed values. There are infinitely many roots, but only finitely many lie to the right of any vertical line. XPP Studio approximates the rightmost ones by Chebyshev collocation of the solution operator [12] and then polishes them with Newton's method. The equilibrium list then reports, for example, *Unstable, 2 roots with Re* $\lambda > 0$ and lists the rightmost roots. **Continue equilibria** uses the same roots, so Hopf points are those of the delay equation.

Try it

1. Load **Delayed logistic equation** and press G. With $r\tau = 1.8 > \pi/2$ the solution settles onto a periodic orbit (Figure 19, left). Change the history formula in the Initial tab and integrate again.
2. The inspector lists two equilibria. $x = 0$ has the single root $\lambda = r$. $x = 1$ has a complex pair $\lambda \approx 0.097 \pm 1.63i$ in the right half-plane.
3. In the Bifurcation view choose `tau`, a range of 0.2 to 1.5 and **Step ds** 0.02, then press **Continue equilibria**. A Hopf point HB1 is found at $\tau \approx 0.874$, against $\pi/(2r) = 0.8727$. **Sweep max/min** adds the oscillation amplitude (Figure 19, right).
4. Load **Mackey–Glass equation** [14] (Figure 21). With $\tau = 17$ the solution is chaotic; the file uses `@ trans=300` to drop the transient. Continuation in τ finds Hopf points at $\tau \approx 4.71$ and 20.98. Linearising by hand gives $\lambda = -0.1 - 0.4\, e^{-\lambda\tau}$, with Hopf points at $\tau = 4.708$ and 20.93.

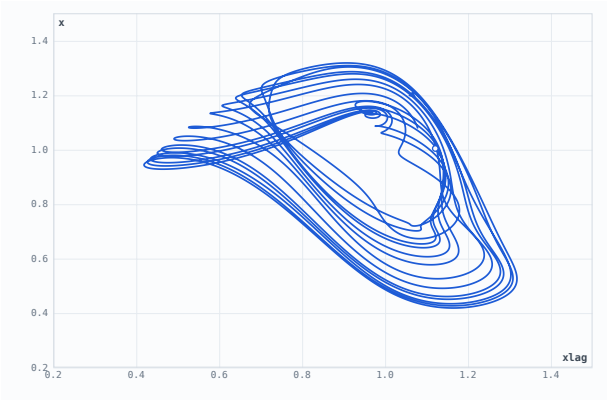


Figure 21: The chaotic Mackey–Glass attractor at $\tau = 17$, drawn as $x(t)$ against $x(t-\tau)$ after a transient of 300.

Periodic-orbit and two-parameter continuation, the direction field and the derivative readout are switched off for delay equations, because the state at time t is a whole function rather than a point.

13 Tutorial 10: events and global flags

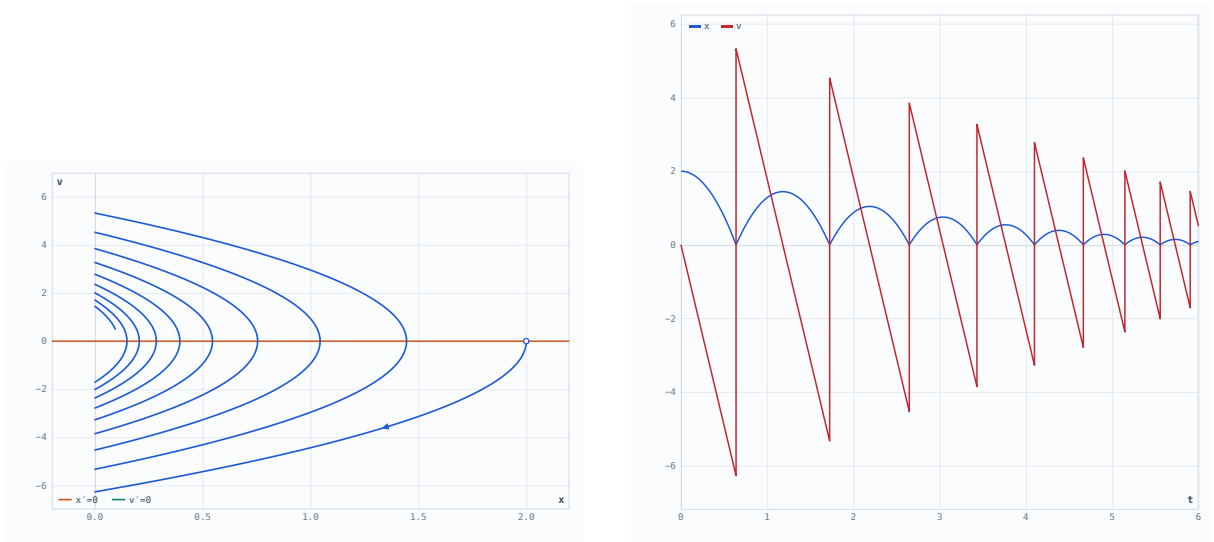


Figure 22: The bouncing ball. Left: velocity against height. Each parabola is one flight, and the line breaks at each event instead of joining the points before and after it. Right: height x (blue) and velocity v (red) against time, with the jump in v at each bounce.

Many models change discontinuously when a threshold is reached: a neuron fires and resets, a ball bounces, a switch flips. XPPAUT describes such events with *global flags*,

```
global sign condition {name1=expr1;name2=expr2;...}
```

The assignments are made when `condition` crosses zero from negative to positive (`sign`$= 1$), from positive to negative (-1), or in either direction (0). All the right-hand sides are evaluated first, then all the assignments are made together. The names may be state or Markov variables, and `arret=1` stops the integration.

How events are located. After every step XPP Studio checks each condition. If one has crossed zero in its direction, the crossing time is found on the cubic Hermite interpolant of that step, by regula falsi on the condition. The point just before the event and the point just after the assignments are both stored, at the same time, and integration continues from there. This works with every integrator, including the adaptive and stiff ones, and with maps, where the assignments are applied to the new iterate.

Try it

1. Load **Bouncing ball**:

   ```
   x'=v
   v'=-g
   global -1 x {x=0;v=-e*v}
   ```

   Press G. The status line reports the number of events. The first two bounces occur at $t = \sqrt{2x_0/g} = 0.6386$ and $t = 0.6386\,(1 + 2e) = 1.7242$, and the computed times agree to six decimal places. Each flight is a parabola in the phase plane (Figure 22). Lower the restitution e with its slider and watch the bounces die away faster.

2. Load **Integrate-and-fire neuron**:

   ```
   v'=(-v+i-w)/tau
   ```

```
w'=-w/tauw
global 1 v-1 {v=0;w=w+b}
```

Each time v reaches threshold it is reset and the adaptation variable w is kicked up, so the firing rate slows until it settles (Figure 23). With $b = 0$ the interspike interval is $\tau \ln(i/(i-1))$, and the computed value agrees to six or more digits.

3. Add `global 1 t-50 {arret=1}` to a model to stop every run at $t = 50$.

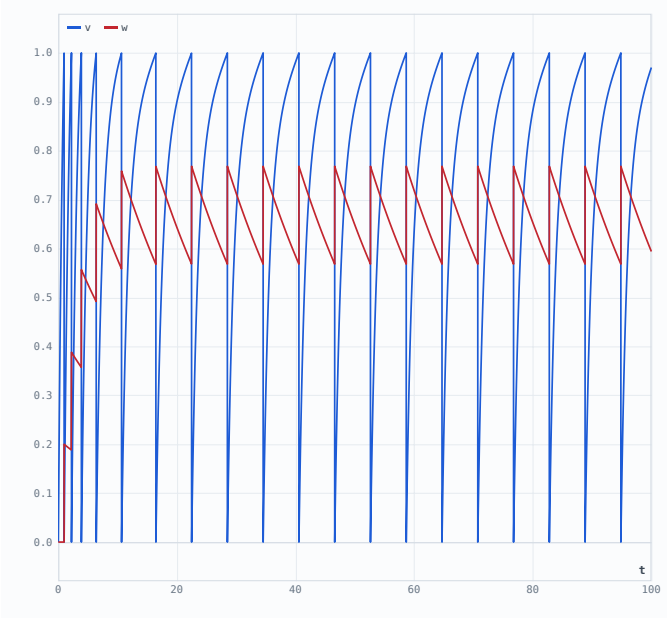


Figure 23: Integrate-and-fire neuron with adaptation: v (blue) is reset to 0 at each threshold crossing, while the adaptation current w (red) builds up and lengthens the interspike interval.

Equilibria, nullclines and equilibrium continuation ignore flags. Periodic-orbit continuation is switched off for models with flags; use **Sweep max/min** for the long-term behaviour instead.

14 Tutorial 11: boundary value problems

XPP Studio also solves two-point boundary value problems by shooting, using XPPAUT's syntax. A line `b` (or `bndry`) followed by an expression gives a condition that the expression vanishes. In it, a plain name `u` is the value at the left end and `u'` the value at the right end. The interval runs from `t0` (`@ t0=`, default 0) to `t0` plus the total time. An unknown parameter, such as an eigenvalue, is written as a variable with zero derivative, `lam'=0`, so that the numbers of unknowns and conditions match. `#@irange u lo hi` gives the initial value of `u` a slider in the Initial tab.

Try it

1. Load **Bratu boundary value problem**:

```
u'=up
up'=-lam*exp(u)
b u
b u'
```

Turn on **Solve BVP** above the plot. Newton's method adjusts the initial values until both conditions hold, and the status line reports the largest residual. Moving the λ

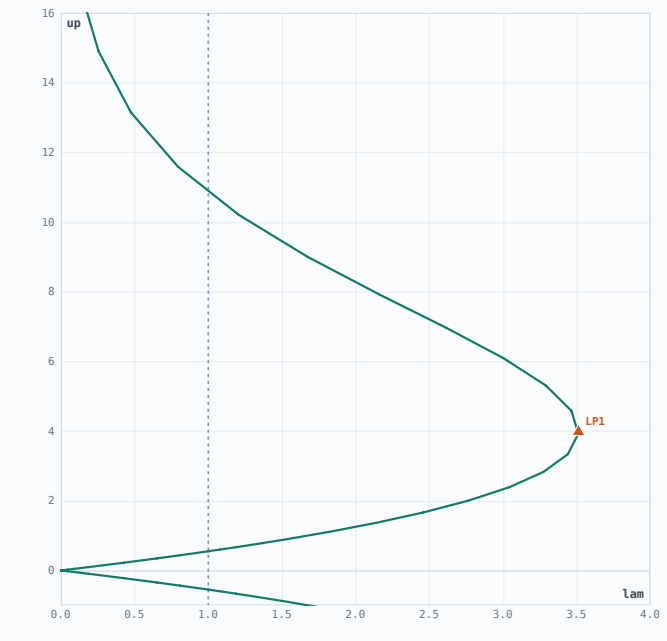


Figure 24: Bratu's problem $u'' = -\lambda e^u$, $u(0) = u(1) = 0$, continued in λ with **Continue BVP**. The vertical axis is the slope $u'(0)$. The branch turns round the fold LP1 at $\lambda \approx 3.5138$ onto the upper solutions.

slider solves the problem again at each new value.

2. In the Bifurcation view set the range of λ to 0–4, choose **Plot** `up`, and press **Continue BVP**. Pseudo-arclength continuation in the initial values and λ follows the branch round the fold at $\lambda \approx 3.5138$ (Figure 24). Clicking the branch loads that solution.
3. The examples **Eigenvalue boundary value problem** and **Rotating spiral wave** have a free parameter written as a variable with zero derivative. The spiral-wave example also starts at $t_0 = 0.01$ to avoid a singular end point.

15 Tutorial 12: scripts for reproducible figures

Each figure in a paper usually comes from a sequence of clicks: load a model, set some parameters, continue a branch, start orbits from a Hopf point, change what is plotted, export. The **Script** tab writes that sequence down as plain text, one command per line, and replays it. The commands call the same functions as the buttons, and each waits for any continuation or grid of trajectories to finish before the next starts. The same script therefore gives the same figures every time, whatever state the window was left in.

Try it

1. Open the **Script** tab and press **Example**. This script is loaded:

```
example ml
par i=60 phi=0.04
view pp
erase
grid
export svg ml-phaseplane.svg
bifpar i 0 300 ds=2
```

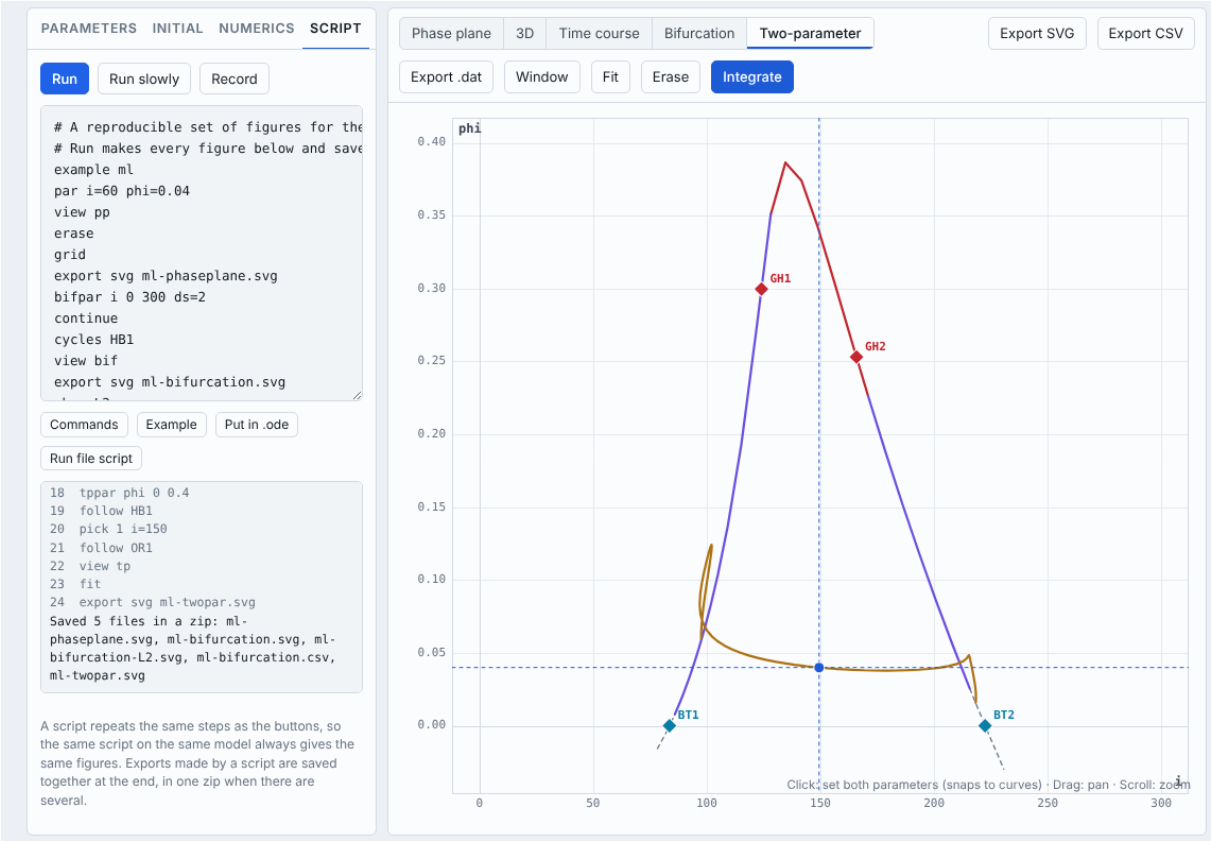


Figure 25: The Script tab after running the example script. The log lists each command as it finished and the files saved; the plot shows the last view the script chose, the two-parameter diagram with the Hopf curve and a curve of constant period through OR1.

```
continue
cycles HB1
view bif
export svg ml-bifurcation.svg
show L2
export svg ml-bifurcation-L2.svg
show hilo
export csv ml-bifurcation.csv
tppar phi 0 0.4
follow HB1
pick 1 i=150
follow OR1
view tp
fit
export svg ml-twopar.svg
```

2. Press **Run** (or Ctrl+Enter in the script). The Morris–Lecar model is loaded, a 10×10 grid of trajectories is drawn, the equilibria are continued in i, the periodic orbits are followed from HB1, the Hopf curve and a curve of constant period are followed in (i, ϕ), and five files are saved together in one zip (Figure 26). It takes about 15 seconds.

3. Change `phi=0.04` to `phi=0.06` and run it again: all five files are remade for the new value.

4. **Run slowly** pauses after each command, so that you can watch the steps. While a

script runs, **Run** becomes **Stop**.

5. **Step** (or Ctrl+Shift+Enter in the script) runs one command and stops. The next line is shown next to the buttons and selected in the script, and the files the command exports are saved straight away. If a command fails, correct the line and press **Step** again: the script can be edited between steps, and stepping carries on from the same line. **Run to end** runs the rest, and **Restart** goes back to the first line.

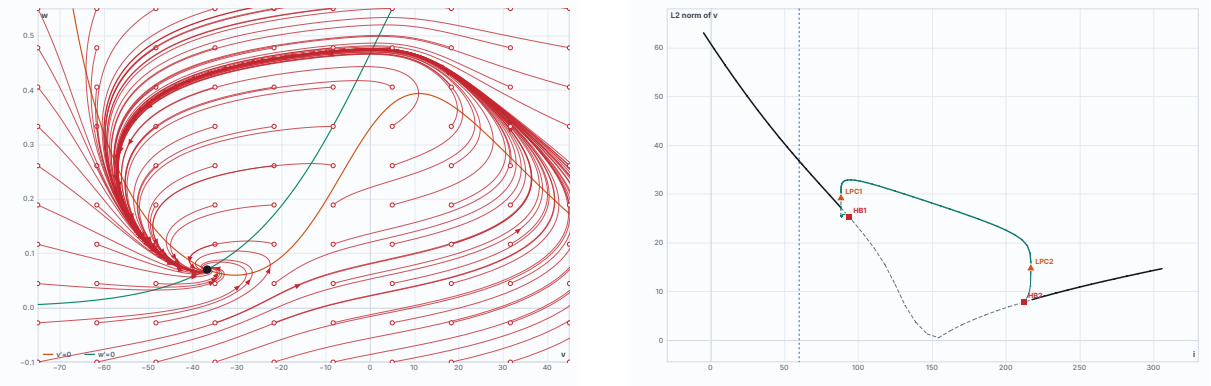


Figure 26: Two of the five figures the example script saves: the phase plane with a 10×10 grid of trajectories, and the bifurcation diagram in i plotted as the L^2 norm of v, with the branch of periodic orbits from HB1 to HB2 and its folds LPC1 and LPC2.

The commands. **Commands** lists them all in the log. Names of variables, parameters and special points are those shown in the interface, and **#** starts a comment.

Command	Effect
`example ml`	load a built-in example (by its id, or part of its name)
`apply`	load the model in the Equations tab
`par i=60 phi=0.04`	set parameters
`init v=-20 w=0.1`	set initial values
`set name=value`	either of the two, whichever the name is
`num total=400 dt=0.05 trans=100`	numerical settings; also `meth`, `dir` (forward, backward or both), `tol`, `bound`, `seed`, continuation settings `ds`, `nmax`, `ntst`, ..., and the continuation tolerances `eqtol`, `potol`, `tptol`, `restol`, `itmax`, `tgrow` (Section 18); `auto` empties a box
`view pp|3d|tc|bif|tp`	choose the plot
`axes v w`	phase-plane axes (three names for 3D)
`window -80 50 -0.1 0.6`	axis limits x_{lo}, x_{hi}, y_{lo}, y_{hi}
`fit`	fit the axes to the data
`nullclines on`, `field off`	also `equilibria`, `points`, `equal`
`tcvars v w`	variables in the time course
`spectrum on`, `spacetime off`	also `logscale`, `smooth`, `legend`, `tp3d`, `solvebvp`
`rotate 30 20 [1.5]`	3D view: azimuth, elevation (degrees) and zoom
`note 12 0.3 [colour=t2] text`	label at (x, y) of the current plot; `\n` starts a new line. The colour is `accent`, `t1`–`t8` or `#rrggbb`
`notes clear`	remove the notes of the current plot
`erase`, `go [both]`	erase; integrate from the initial values
`traj v=-40 w=0.2 [backward]`	a trajectory from a given point
`grid`, `grid v -60 40 5 w 0 0.5 5`	grid of initial conditions (default 10×10 over the window)
`separatrices [k]`	separatrices of the kth saddle
`findeq`	search the window for equilibria
`bifpar i 0 300 ds=2`	continuation parameter, range and step
`continue [eq k]`	continue equilibria
`cycles HB1`, `double PD1`	orbits from a Hopf point; period-doubled orbits (flows and maps)
`switch BP1`	the other branch through a branch point
`period 3`	maps: continue period-3 orbits (also `bifpar ... period=3`)
`limitcycle`, `sweep`	continue the orbit reached by the trajectory; sweep the range
`plot v`, `show L2`	variable, and hi-lo, L2, period or frequency
`pick 1 i=150`	pick an orbit of family 1 (it becomes OR1, OR2, ...)
`tppar phi 0 0.4`, `follow HB1`	second parameter; follow a special point (OR1: constant period)
`slices HB1 gca 3 5.5 6`	follow HB1 at 6 values of a third parameter
`connect 2`, `connect 3 1 [backward]`, `connect end`	find a homoclinic orbit of equilibrium 2, or a heteroclinic orbit from 3 to 1; `end` is the saddle found at the end of the last branch of orbits (Section 17). `follow HOM1` follows it
`clear bif`, `clear tp`	clear a diagram
`export svg|csv|dat [file]`	save the current view or its data
`writestate`, `echo`, `wait ms`	write the state into the file; note; pause
`history save [file]`, `history clear`	save the status-line history as a text file (with the other exports), or empty it

If a command fails (an unknown name, a Hopf point that is not there, a continuation that does not start), the script stops and the log gives the line number and the reason. Files exported before then are still saved.

Recording. You need not type a script. Press **Record**, then work as usual: choosing a model, moving sliders, changing settings, integrating, continuing, following curves and exporting are written into the script as you go. A slider dragged across many values gives a single `par` line with its final value. Actions are recorded however you start them, from a button, a row of the special-points table, the equilibrium list or a key, and from any view, so **Follow** or **Cycles** in the table gives the same lines as the buttons in the inspector. What you do with the mouse is recorded too:

- clicking the phase plane gives a `traj` line with the starting point (and `both` for a shift-click);
- panning and zooming give a `window` line once the gesture ends, and rotating or zooming the 3D view a `rotate` line, so the figure is framed as you left it;
- clicking a diagram to load a point (an equilibrium, a point on a boundary value branch or a two-parameter curve) gives a `set` line with the parameter values and all the initial values it loads, and clicking an orbit branch gives `pick`;
- **Find** for a homoclinic or heteroclinic orbit gives a `connect` line, and **Follow** on HOM1 or HET1 a `follow` line;
- placing a label with **Note** gives a `note` line with its position, colour and text. Moving or editing the label afterwards is not recorded, and neither is its size;
- switching views, by button or key, gives a `view` line, and the chips above the plot (spectrum, space-time, legend, 3D) and the variables chosen for the time course are recorded as well.

Repeated changes of the same thing (dragging a slider, panning in several strokes) give a single line with the final value. Press **Record** again to stop, then edit the result as you like.

If the script box already holds commands when you press **Record**, you are asked whether to clear them or to keep them and start a new block, marked by a dated comment line. A script belongs to a model: loading a different model clears the box, and **Restore old script** brings the old one back.

Scripts inside `.ode` files. **Put in .ode** copies the script into the equations as lines starting `#@do`, just before `done`:

```
#@do par i=60
#@do bifpar i 0 300
#@do continue
#@do cycles HB1
#@do view bif
#@do export svg bifurcation.svg
done
```

XPPAUT reads these as comments. **Run file script** in the Script tab reloads the model and replays them, so a single `.ode` file can carry both a model and the figures made from it.

16 Tutorial 13: a forced oscillator and Arnold tongues

In a periodically forced oscillator, or in a phase model, the state lives on a torus. The **Forced phase oscillator** example has two phases, x for the oscillator and $y = \omega t$ for the forcing:

```
p(x)=-sin(x)/(1+a*cos(x))
x'=1+b*p(x)*cos(y)
y'=om
par om=2,b=.25,a=.25
@ fold=x,fold=y,tor_per=6.283185
```

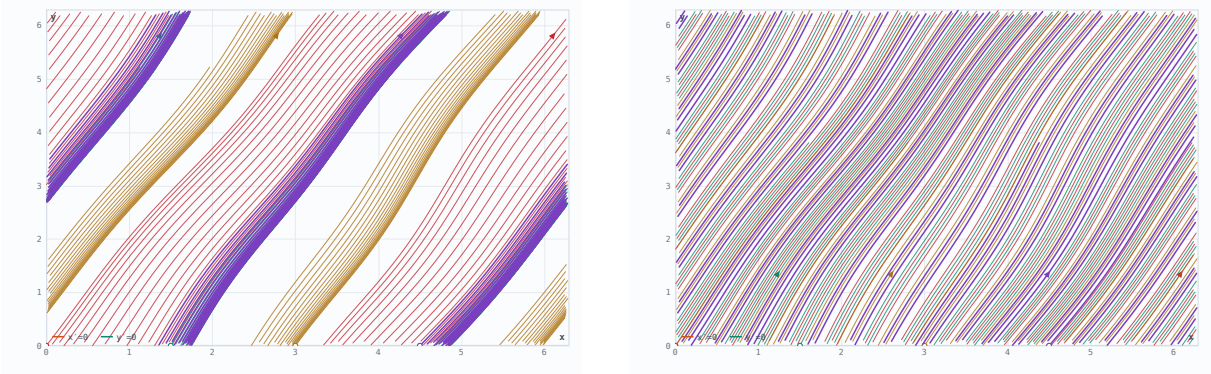


Figure 27: The forced phase oscillator on the torus, x against y (both reduced modulo 2π, so the lines break where they wrap). Left: $\omega = 2$, inside the 1:2 tongue; the trajectories from several starts all close in on the same band, the locked orbit. Right: $\omega = 2.3$, outside the tongue; the lines do not settle onto a closed orbit.

Both variables are declared periodic with `@ fold=`. Plots break where they wrap, and the closing condition of a periodic orbit is taken modulo the period. A locked solution closes in that sense: $x(T) = x(0) + 2\pi n$ and $y(T) = y(0) + 2\pi m$ with $T = 2\pi m/\omega$. Such orbits are followed by the same continuation as limit cycles. The set of parameters for which a locked solution exists is an *Arnold tongue*, and its two edges are folds of cycles.

Try it

1. Load **Forced phase oscillator** and press G with $\omega = 2$. The trajectory settles onto a locked orbit in which x goes once round while the forcing goes twice (Figure 27, left). Change ω to 2.3 and integrate again to see that it is no longer locked.
2. Return to $\omega = 2$, integrate, and open the Bifurcation view. Set the range of `om` to 1.2–3 and press **Limit cycle**. The locked orbit exists only for ω between 1.9784 and 2.0416: there are two folds of cycles, LPC1 and LPC2, the edges of the 1:2 tongue. The stable and the unstable locked orbit meet at the two folds, so the branch is a closed loop. The continuation stops when it has gone once round, and the status line says that the branch closed up on itself.
3. Open the Two-parameter view, set the range of `b` to 0–0.5, and press **Follow** on LPC1 and on LPC2 in the Special points table. The two curves are the edges of the tongue in (ω, b), and they meet at $b = 0$, $\omega = 2$.
4. To add the 1:1 tongue, press **Clear diagram** in the Bifurcation view. This erases the one-parameter branch and leaves the two-parameter curves. Set $\omega = 1$, integrate, set the range of `om` to 0.5–3, continue in `om` with **Limit cycle** (the folds are at 1.1233 and 0.8684), and follow both folds. Turn on **Legend** and **Colours** (Figure 28).

The tongue is the region between the two edges, widening as the forcing b grows. Narrower tongues, such as 1:3, are harder to reach: at $\omega = 3$ the default run does not settle onto a locked orbit, so you need a longer integration or a starting point on the orbit. One-parameter diagrams add up in the same way: continuing the orbit in ω at three values of b keeps three branches on the same axes, and **Legend** tells them apart by the value of b.

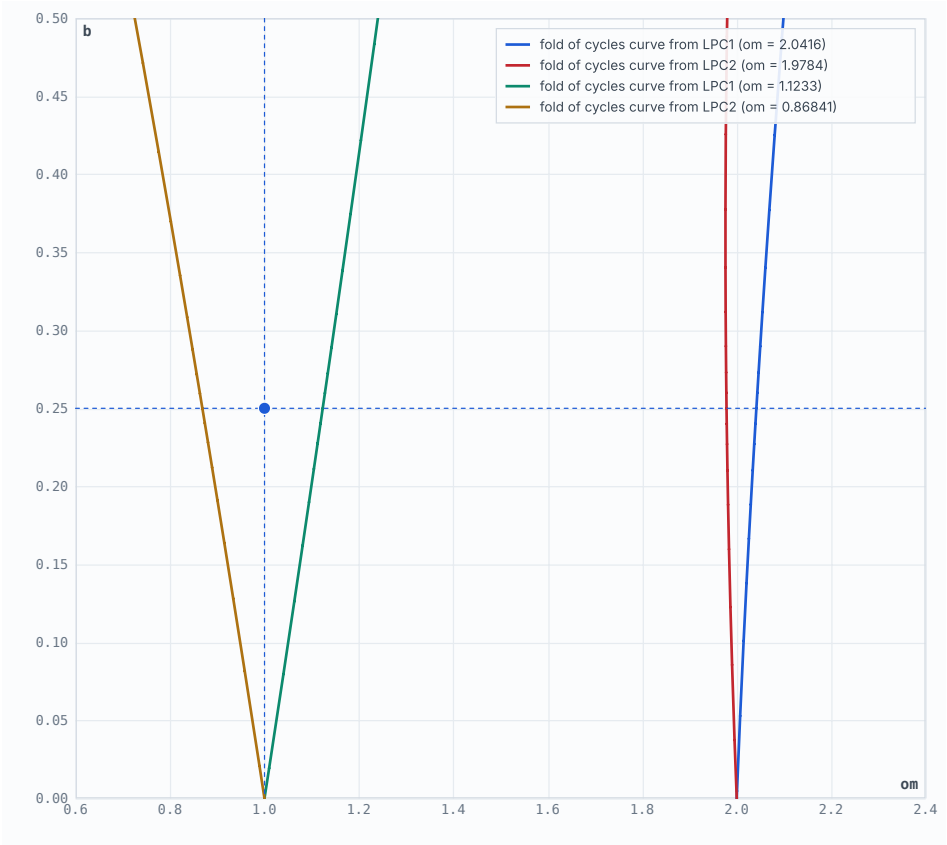


Figure 28: The 1:1 and 1:2 Arnold tongues of the forced phase oscillator in (ω, b), as curves of folds of cycles. Each tongue has two edges, and the legend says which fold and which value of ω each curve was started from. The dashed lines mark the current parameter values.

17 Tutorial 14: homoclinic and heteroclinic orbits

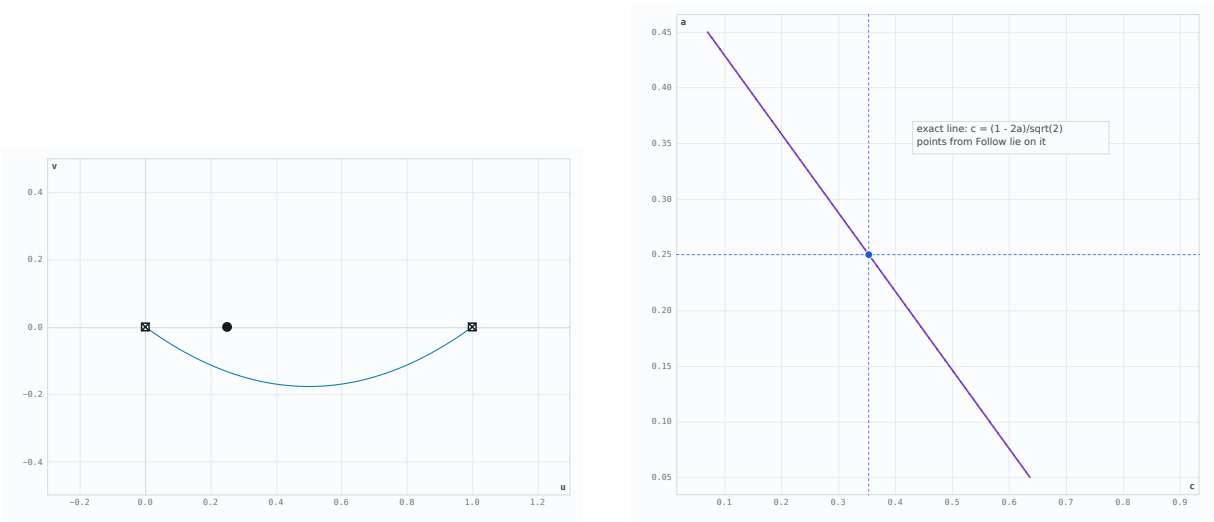


Figure 29: The Nagumo front. Left: the heteroclinic orbit in the (u, v) plane, from the saddle at $u = 1$ to the saddle at $u = 0$ (squares), with the focus at $u = a$ (dot). Right: the speed c at which it exists, followed in a from 0.05 to 0.45. The points lie on the exact line $c = (1 - 2a)/\sqrt{2}$ to about 10^{-9}.

A *homoclinic orbit* leaves a saddle and comes back to it; a *heteroclinic orbit* joins two saddles. Both exist only for special values of a parameter: they are codimension one, so a one-parameter

diagram shows them as single points where a family of orbits ends. They matter for waves. Written in a frame moving with speed c, a travelling front of a reaction–diffusion equation is a heteroclinic orbit of an ordinary differential equation between two rest states, and a pulse is a homoclinic orbit. The speed c is the parameter that has to be tuned until the connection exists.

How the orbit is found. The orbit starts a small distance from the first saddle, along the eigenvector of its single unstable eigenvalue, and is integrated until it comes closest to the second. There the program measures on which side of the second saddle's stable manifold it passes, using the unstable eigenvector of the second saddle. The measure changes sign at a connection. XPP Studio scans the range of the continuation parameter for a sign change, narrows it down by false position (the Illinois variant, which needs far fewer integrations than bisection), and accepts the result only if the orbit really goes on to the target. **Follow** then steps a second parameter and solves again for the first near the previous value, in both directions.

The controls are in the section **Homoclinic and heteroclinic orbits** of the Bifurcation panel, under the continuation buttons. The menus **From** and **To** list the equilibria found in the phase plane at the current parameters, and any saddle found at the end of a branch of periodic orbits (below). **Find** searches the range of the continuation parameter, reporting its progress in the status line. While it searches, **Find** becomes **Stop**: press it, or Esc, to end a search that is taking too long. **Follow** can be ended the same way: Esc moves on to the other direction, and **Stop** in the Two-parameter view (or Shift+Esc) ends it, keeping the points found so far. The orbit found is drawn in the phase plane and becomes a point of the diagram, HOM1, HET1, ... in the table of special points. On the diagram it is marked by diamonds at the maximum and minimum of the plotted variable along the orbit, at its L^2 norm in the L2 view (for a homoclinic orbit, the norm of the saddle, the limit of the norms of periodic orbits that approach it), at zero frequency, and by a dashed line in the period view, where the period is infinite. Clicking its row moves to it and draws it again. **Follow** in its row continues it in Parameter 2 of the Two-parameter view, over the range set there; the curve is drawn in the Two-parameter view in its own colour, and HOM1 and HET1 also appear under **Start** there.

Try it

The Nagumo front.

1. Load **Nagumo front (heteroclinic orbit)**. The equations are $u' = v$, $v' = -cv - u(1-u)(u-a)$, with the two saddles at $u = 0$ and $u = 1$ and a focus at $u = a$.
2. Open the Bifurcation view; the continuation parameter is c. Under **From** choose the saddle at $u = 1$, under **To** the saddle at $u = 0$.
3. Press **Find**. At $a = 0.25$ the speed is $c = 0.353553392$, the exact value $(1-2a)/\sqrt{2}$ to the digits shown. The phase plane shows the orbit (Figure 29, left), and HET1 is added to the special points.
4. In the Two-parameter view, Parameter 2 is a. Press **Follow** on HET1 in the special points table (or choose HET1 under **Start** and press **Follow curve**). In a second or two the curve of speeds over $0.05 \le a \le 0.45$ appears (Figure 29, right), and a note placed on it (Section 2) can carry the formula.

Try it

A homoclinic orbit of Morris–Lecar, found from the periodic orbits. The Hopf-regime example has no saddle, so a homoclinic orbit cannot exist there. The example **Morris–Lecar homoclinic orbit (class 1 parameters)** has three equilibria at $i = 37$: a stable

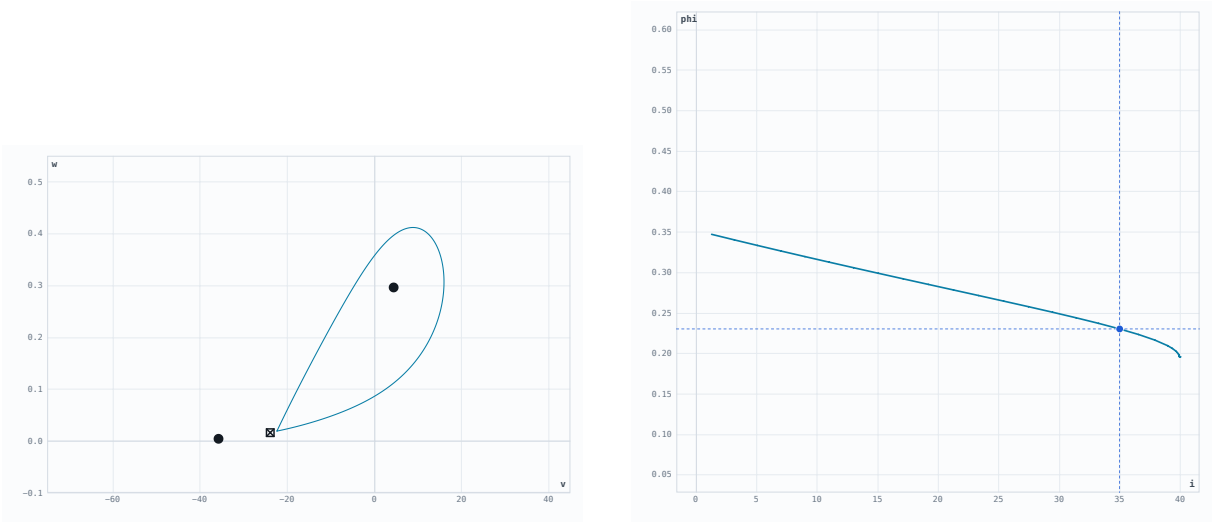


Figure 30: A homoclinic orbit of the Morris–Lecar model with $v_3 = 12$, $v_4 = 17.4$, $g_{Ca} = 4$. Left: at $\phi = 0.23$ the orbit leaves the saddle (square), goes round the focus (upper dot) and returns to the saddle at $i = 35.0067343$. The lower dot is the stable node. Right: the homoclinic curve in (i, ϕ), followed in ϕ. It ends where the saddle merges with the node, near $i = 40$, and where it reaches the left edge of the range.

node, a saddle and a stable focus.

1. Load it and open the Bifurcation view, with the parameter i over its range 0 to 40. Choose the stable focus under **Start** and press **Continue equilibria**: the branch has a subcritical Hopf point HB1 at $i = 36.316$.

2. Press **Cycles** on HB1. The unstable orbits grow, turn at the fold LPC1 ($i = 40.593$) and come back, and their period grows without bound as i approaches 35.0068: the branch stops when the period reaches 40 times its starting value (**Period growth stop**, Section 18).

3. Whenever a branch of periodic orbits ends with its period still growing, XPP Studio looks at its last orbit. Where the orbit moves most slowly it passes close to an equilibrium; Newton's method from that point finds it, and if it is a saddle with one real unstable direction (or one real stable direction, with time running backwards) it is loaded under **From** and **To**, as *end of HB1 orbits: saddle at ...*, and the status line says so. Here it is the saddle at $v = -22.3$.

4. Press **Find**. A narrow window round the end of the branch is searched first, and in a second or two the status line reports a homoclinic orbit at $i = 35.0067343$. HOM1 appears at the end of the branch of orbits (Figure 31), and the orbit in the phase plane (Figure 30, left).

5. In the Two-parameter view set Parameter 2 to ϕ (range 0.05 to 0.6), and press **Follow** on HOM1. The curve in the (i, ϕ) plane (Figure 30, right) takes under a minute. It falls from $\phi \approx 0.35$ at small i to $\phi \approx 0.2$ near $i = 40$, where the saddle and the node merge and the connection is lost.

The saddle can also be chosen by hand: it is in the **From** and **To** menus as one of the equilibria of the phase plane.

When to run time backwards. The method needs the starting saddle to have exactly one real unstable direction, and the target saddle too, with all the other directions stable. If instead

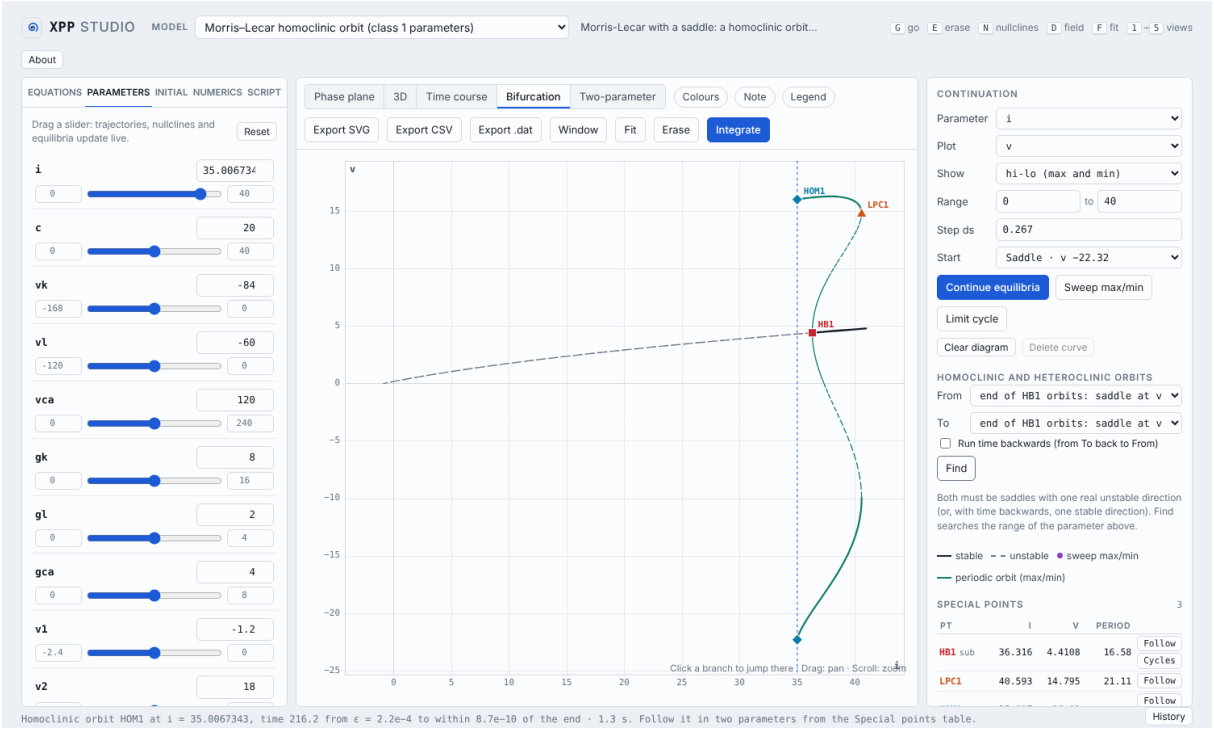


Figure 31: The Bifurcation view after **Cycles** on HB1 and **Find** for the Morris–Lecar homoclinic orbit. The branch of periodic orbits from HB1 turns at LPC1 and ends, with its period growing, at the homoclinic orbit HOM1, marked by two diamonds at the maximum and minimum of v. The section **Homoclinic and heteroclinic orbits** in the right-hand panel holds the saddle found at the end of the branch, under **From** and **To**.

the starting saddle has a one-dimensional *stable* manifold and a bigger unstable one, as can happen at the rest state of a pulse, tick **Run time backwards**. The orbit is then followed from **To** back to **From**, which turns the one-dimensional stable manifold into the starting direction.

Limits.

- Both equilibria must be saddles with one real unstable direction (counted backwards in time if **Run time backwards** is ticked). Saddles with complex unstable eigenvalues, or with a two-dimensional unstable manifold at both ends, are not handled.
- **Follow** steps Parameter 2 and solves for Parameter 1, so a curve that turns back in Parameter 2 is lost at the turn. The status line says whether the curve ended at the edge of a range or because the connection was lost.
- The method has been tested on planar problems, where the answers above are exact or checked. For systems with three or more variables, check an orbit by integrating it, and do not rely on a failure to find one as proof that none exists: the range of Parameter 1 and the choice of equilibria matter.
- The orbit is not tested for stability, and codimension-two points are not located on the curve. Orbits found are not saved in the `.ode` file, but a script can find them again with `connect` and follow them with `follow` (Section 15).
- The saddle at the end of a branch is only looked for when the period is still rising over the last few orbits and has at least doubled along the branch. If the branch stopped for another reason first (the point limit, say), raise the limit and continue again, or choose the saddle by hand.

18 Numerical settings

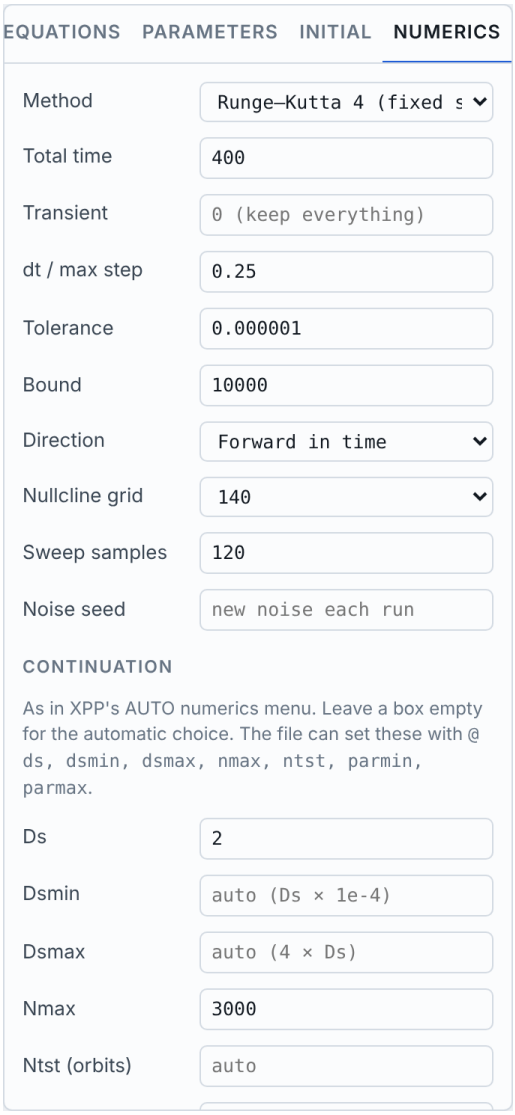


Figure 32: The Numerics tab: integration settings at the top, continuation settings below, and the continuation tolerances (Section 18) further down.

The **Numerics** tab (Figure 32) collects the integration and continuation parameters. Changes to the integration settings rerun every trajectory on screen.

Integration.

Method As in Section 3.

Total time The duration of each run; for maps, the number of iterations.

Transient As in XPPAUT, the run still lasts the total time, but only $t \geq$ the transient is kept (`@ trans=`). Use it to look at attractors without the approach to them. Manifolds and the single periods loaded from continuation are never cut.

dt The step for fixed-step methods, and the largest step for adaptive ones.

Tolerance Applies to the adaptive methods.

Bound Integration stops when any variable exceeds this value.

Direction Forward, backward or both ways in time (also next to **Integrate** in the Initial tab).

Start time The start time t_0 of each run (`@ t0=`).

Nullcline grid Sets the resolution of the nullclines.

Sweep samples Sets the number of parameter values in **Sweep max/min**.

Continuation. These settings correspond to the entries of XPPAUT's AUTO numerics window and can be set in the file with `@ ds, dsmin, dsmax, nmax, ntst, parmin, parmax`. Leave a box empty to use the automatic choice.

Setting	Effect
Ds	initial arclength step for equilibrium branches (same as **Step ds**)
Dsmin, Dsmax	smallest and largest steps; a branch stops if the step falls below Dsmin
Nmax	maximum number of points in each direction of an equilibrium branch
Ntst (orbits)	number of shooting intervals per periodic orbit (automatic: 20 for up to 3 variables, 14 up to 5, otherwise 10)
Ds, Dsmax, Nmax (orbits)	the same for periodic-orbit branches and curves of cycles
Nmax (2-par), Nmax (2-par cycles)	point limits for two-parameter curves

Continuation tolerances. Each step of a continuation predicts the next point along the tangent and corrects it with Newton's method. The correction stops when its size falls below a tolerance, and the point is then accepted only if the equations hold to within the residual tolerance; otherwise the step is halved. The section **Continuation tolerances** of the Numerics tab sets these, and **Defaults** puts them back. The file can set them with `@ eqtol, potol, tptol, restol, itmax, tgrow`, and **Write state** writes those that differ from the defaults.

Setting	Default	Effect
Newton tol (eq.)	10^{-10}	last Newton correction for branches of equilibria and fixed points, relative to the size of the solution (`eqtol`, like AUTO's EPSU)
Newton tol (orbits)	10^{-9}	the same for periodic orbits, in the scaled unknowns of the shooting method (`potol`)
Newton tol (2-par)	10^{-8}	the same for two-parameter curves of folds, Hopf points and bifurcations of cycles (`tptol`)
Residual tol	10^{-6}	largest residual of the equations at an accepted point; for orbits, relative to their size (`restol`)
Newton iterations	automatic	most Newton iterations per step before the step is halved: 15 for equilibria, 14 for orbits and two-parameter curves, 12 for curves of cycles (`itmax`, like AUTO's ITNW)
Period growth stop	40	a branch of periodic orbits, or a curve of cycles, stops when the period reaches this multiple of its starting value (`tgrow`)

Tighter tolerances give more accurate points and usually smaller steps; looser ones can carry a branch through a stretch where the corrector struggles, at some cost in accuracy, so check such a branch against a tighter run. Raise the period growth stop to follow orbits closer to a homoclinic orbit.

If a branch stops early, the status line says why: for example, it reached the point limit, the corrector stopped converging, the orbit shrank onto a Hopf point, or the period grew without bound (a homoclinic orbit is near). If it stopped because of a step-size or point limit, adjust the corresponding setting and run it again. A branch of orbits whose period grew to its end also loads the saddle it approaches into the homoclinic search (Section 17).

19 Saving and exporting results

Export SVG Saves the current view as a vector figure with light-theme colours, ready for a paper or for editing in Inkscape or Illustrator. Legends and notes are included. All the plots in this document were made this way.

Export CSV Available in the Time course, Bifurcation and Two-parameter views.

- The time-course table has every trajectory on screen, one row per stored point, with the trajectory number, its kind (trajectory or manifold), its direction (forward or backward), t and every variable.
- The bifurcation table has one row per computed point, with these columns: curve, kind (equilibrium, periodic orbit, sweep or special), label (HB1, LPC2, ...), parameter name and value, stability, number of unstable directions, period, the state, and the maximum and minimum of each variable along an orbit.
- The two-parameter table has the two parameter values, the period where relevant, the state, and labelled rows for codimension-two points. Hopf curves also get columns for ℓ_1 and for criticality (subcritical, supercritical or neutral saddle).
- In the bifurcation table, special rows for Hopf points give ℓ_1, and the stability column says subcritical or supercritical.

Export .dat In the Bifurcation and Two-parameter views, saves the same continuation data as `.dat` files: numbers only, no header and space-separated columns, the form XPPAUT and gnuplot read. The bifurcation diagram gives one file each for equilibria, periodic orbits and sweeps. The status line lists the columns, and text fields become numeric codes.

Copy CSV, Save .dat In the trajectory list, copy the current trajectory (time, variables and auxiliaries) to the clipboard, or save it as a `.dat` file.

Copy text Copies the `.ode` file, including any edits, so that it can be saved or used in XPPAUT.

Writing the current state into the file

After exploring a model you will often want to keep the parameter values and settings you have arrived at. **Write state**, in the Equations tab, writes them into the `.ode` text, much as XPPAUT does when it saves a file:

- parameter values and initial conditions are changed where they are defined, in the `par`, `init` and `x(0)=` lines, as are slider ranges in `#@range` lines and history formulas of delay equations;
- the integration settings (method, total time, step, transient, tolerance, bound, random seed), the plot axes and window, and the continuation settings are collected in `@` lines just before `done`.

For example, after setting $i = 150$ and a transient of 100 in the Morris–Lecar example, the end of the file reads

```
param i=150,c=20
...
# Settings written by XPP Studio, 2026-10-20 10:30
@ meth=rungekutta, total=400, dt=0.25, trans=100, bound=10000
@ xp=v, yp=w, zp=ica, xlo=-75, xhi=45, ylo=-0.1, yhi=0.55
@ ds=2, nmax=3000, parmin=0, parmax=300
done
```

Pressing **Write state** again replaces this block rather than adding another, and **Apply** on the result restores exactly the same state. The text is not reloaded, so the trajectories on screen are kept. Use **Copy text**, or your browser's own storage of the editor, to keep the file. Parameters defined through array lines are left as written.

20 Differences from XPPAUT

XPP Studio aims to read the same files and to feel familiar, but it is a reimplementation:

- **Continuation.** It is built in and written from scratch: pseudo-arclength continuation of equilibria, and multiple shooting (rather than AUTO's orthogonal collocation) for periodic orbits. Results agree with AUTO to plotting accuracy on the examples, but very long or very unstable orbits may need more shooting intervals (Ntst).
- **Delay equations** are integrated with a fixed step only, and periodic orbits of delay equations cannot be continued. XPPAUT's other delay options, such as `delshft` and `ndelay`, are not supported. The `@ delay` option is accepted but not needed, since all of the history is kept.
- **Global flags** can set state and Markov variables and `arret`. The XPP output flag `out_put` is accepted and ignored.
- **Not supported:** `special` and `export` declarations, and `set` blocks.
- **Two-parameter Hopf curves** are limited to 12 variables, because the bialternate product grows quadratically with dimension.
- **Homoclinic and heteroclinic orbits** (Section 17) are found by shooting along the one-dimensional unstable manifold of a saddle, not by AUTO's HomCont projection boundary conditions, so they are limited to saddles with one real unstable direction (or one real stable direction, with time running backwards).
- **Volterra models** cannot be continued, since their state is infinite-dimensional.
- **Browser files.** Files are opened and saved through the browser. Table data files are kept in the browser's local storage until you press **Forget table files**.

21 Credits and licence

XPP Studio is inspired by, and reads the model files of, XPPAUT by G. Bard Ermentrout (University of Pittsburgh), described in [1]. Its continuation methods follow the ideas of AUTO [2] by Eusebius Doedel and colleagues, and of MatCont [3]. This version (October 2026) was created by Peter Ashwin and Bard Ermentrout in collaboration with Claude Opus 5.5, an AI model made by Anthropic.

XPP Studio is free software, released under the GNU General Public License, version 3 or later. The licence text is distributed with the program (`LICENSE.txt`) and is available at `https://www.gnu.org/licenses/gpl-3.0.html`.

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